Generalized AKM: Theory and Evidence*

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July 2026

Abstract

Standard AKM wage decompositions assume observed covariates enter log wages linearly. We develop a semiparametric leave-out estimator that replaces the linear control function with an unknown smooth function, approximated by a growing polynomial basis. We prove consistency and asymptotic normality with heteroskedastic errors and a number of fixed effects that grows proportionally with sample size. Quadratic variance components require stronger smoothness than linear functionals do—the unknown function must have more derivatives than continuous covariates—but the condition is mild in wage applications. In Portuguese employer–employee data for 2008–2018, adding worker and firm-input controls in a linear additive specification lowers worker-effect variance from 0.553 to 0.478 of total wage variance, firm-effect variance from 0.136 to 0.115, and sorting from 0.088 to 0.053. Replacing this specification with common or group-specific nonlinear bases changes the firm and sorting components only modestly; worker-effect variance is more sensitive to the basis used for group-specific controls.

Keywords: wage decomposition, many regressors, semiparametric model, series estimator
JEL codes: J31, J62, C14, C23, C55

*We are grateful to Edoardo M. Acabbi, Stanislav Anatolyev, Andrea Alati, Matias D. Cattaneo, Patrick Kline, Michal Kolesár, Mikkel Plagborg-Møller, Ulrich Müller, and seminar participants at Princeton University, XXIII Brucchi Luchino conference, and IRLE Berkeley for fruitful comments and suggestions. This paper results from research funded under the umbrella of the ERC-CZ project No. LL2319. Funding from the European Union's Horizon 2020 research and innovation program under the Marie Skłodowska-Curie grant agreement No. 870245 is also gratefully acknowledged.

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1. Introduction

Decomposing wage dispersion into worker and firm components is a basic question in labor economics. Since Abowd, Kramarz, and Margolis (1999), the standard approach has been to estimate a two-way fixed-effects model—commonly known as AKM—in matched employer–employee data and report how much of the overall wage variation is attributable to workers, to firms, and to their sorting. A large literature has refined the statistical foundations of this decomposition, including the leave-out bias correction of Kline, Saggio, and Sølvsten (2020), which delivers valid inference under heteroskedasticity and many fixed effects. Throughout, however, both the original estimator and its modern refinements assume that observed covariates—age, education, occupation, and increasingly firm-side variables such as employment, fixed assets, or intermediate inputs—enter log wages through a known linear function. This is hard to reconcile with evidence that returns to experience and schooling are nonlinear (Mincer 1974; Card et al. 2018), and there is no strong reason to expect firm-level controls to be additively separable from worker characteristics. Whether and how this linearity restriction distorts the decomposition in practice is an open question.

This paper develops a semiparametric estimator for variance components in models where covariates enter through an unknown function. Correctly attributing wage variation to workers, firms, and their sorting requires isolating the role of observed characteristics from unobserved heterogeneity; if the functional form linking covariates to wages is misspecified, the decomposition inherits the distortion. The model retains the two-way fixed-effects structure of AKM but replaces the linear control function with an unspecified smooth function, approximated by a growing polynomial or spline basis. Our estimator combines the leave-one-out bias correction of Kline, Saggio, and Sølvsten (2020) with this flexible approximation: the standard plug-in decomposition is corrected using leave-one-out residual variances computed from the augmented regressor set that includes both the high-dimensional fixed effects and the series basis. We show that this estimator is consistent and asymptotically normal for the original variance components under heteroskedasticity, a growing number of fixed effects, and an explicit smoothness condition on the unknown function.

The main theoretical challenge is that three problems must be handled jointly: many-regressor bias from high-dimensional fixed effects, heteroskedasticity-robust variance estimation, and approximation error from the unknown control function. Existing leave-out estimators solve the first two but assume linearity; existing semiparametric series theory handles the third but applies to linear functionals of the coefficients, not quadratic forms. Because variance components are quadratic in the coefficient vector, approximation error enters both the plug-in estimator and the leave-out bias correction, and the two sources must vanish jointly. This imposes a smoothness requirement on the unknown function that is stronger than what standard semiparametric estimation of regression coefficients demands: the function must have more continuous derivatives than the number of continuous covariates entering flexibly. In wage applications, where those covariates are quantities such as age, firm employment, fixed assets, or intermediate inputs, the condition is mild. We prove consistency, a chi-squared limit when the variance component has fixed rank, and a normal limit for the empirically relevant case where the rank grows with sample size. A random-projection approximation extends the estimator to the large-scale administrative datasets typical of this literature.

The simulation study shows when the added flexibility matters. Under strong nonlinearity, higher-degree polynomial bases reduce signed bias, especially when misspecification interacts with high leverage. When the true relationship is linear or mildly nonlinear, all estimators perform similarly, so the additional flexibility imposes little cost when it is not needed. The simulations also show that finite-sample inference is harder than point estimation: point estimates remain stable across the designs, but the nominal confidence intervals remain somewhat below 95% coverage in several stress cases, most clearly under high leverage. In a calibration that holds the Portuguese worker–firm graph fixed, the leave-out (KSS) correction has point-estimate bias below 0.2% in all four nuisance designs, against 1.2–1.6% for the homoskedastic correction and 1.9–2.4% for the plug-in estimator. For applied work, the estimator is therefore most reliable for the decomposition itself. The confidence intervals are best read as diagnostic rather than exact when leverage is high or the residual variances $\hat{\sigma}_i^2$ are hard to estimate.

We use Portuguese linked employer–employee administrative data for 2008–2018 to study the same sequence of increasingly flexible specifications on a common leave-match-out sample. In the specification with education, qualification, and firm inputs, adding the controls in a linear additive specification lowers worker-effect variance from 0.553 to 0.478, firm-effect variance from 0.136 to 0.115, and sorting from 0.088 to 0.053. A common degree-5 polynomial yields corresponding components of 0.495, 0.113, and 0.048; interacting that basis with worker groups yields 0.493, 0.110, and 0.049. Hermite produces nearly the same Panel B decomposition, while the group-specific cubic B-spline yields worker and firm variances of 0.476 and 0.107, with sorting at 0.048. In the parsimonious panel with only gender and age, all five specifications produce similar decompositions. The largest empirical movement therefore comes from adding observed controls. Functional-form flexibility produces smaller changes in firm variance and sorting, while the worker component is more basis-sensitive.

Contribution to the literature. The paper contributes to three strands of the literature. First, it extends the leave-out variance-component estimator of Kline, Saggio, and Sølvsten (2020) from linear many-regressor models to semiparametric models with a growing approximation space, providing the conditions under which leave-out validity survives flexible approximation of the control function. Second, it complements the semiparametric series literature (Donald and Newey 1994; Cattaneo, Jansson, and Newey 2018a, 2018b) by moving from slope coefficients to variance components, identifying the stronger smoothness condition this shift requires. Third, it contributes to the empirical literature on the sensitivity of AKM decompositions (Andrews et al. 2008; Bonhomme et al. 2023; Bonhomme, Lamadon, and Manresa 2019) by separating the movement due to adding observed controls from the smaller, basis-dependent movement due to relaxing their functional form. Our approach differs from grouped-firm or latent-type models (Bonhomme, Lamadon, and Manresa 2019) in that it preserves the original variance components and asks whether leave-out estimation remains valid once the control function is flexibly specified, rather than replacing the AKM structure with a different earnings model.

Outline. Section 2 introduces the semiparametric model and formalizes the variance components. Section 3 derives the estimator and its finite-sample representation. Section 4 establishes consistency, summarizes the strengthened smoothness condition, and collects the core rate restrictions used throughout the theory. Section 5 presents the random projection approximation. Section 6 derives the fixed-rank and growing-rank limiting distributions of our proposed estimator. Section 7 reports our simulation exercise, while Section 8 illustrates an empirical application on linked employer-employee data. Section 9 concludes.

2. Setup

We consider the following semiparametric model,

$$y_i = x_i'\beta + f(z_i) + e_i, \quad i = 1, \dots, n, \tag{1}$$

where the regressors $x_i \in \mathbb{R}^p$, $z_i \in \mathbb{R}^d$ are non-random. The unknown function $f(z)$ belongs to the class of smooth and real-valued functions, $f \in \mathcal{F}$. The unobserved errors $\{e_i\}_{i=1}^n$ are mutually independent and obey $\mathbb{E}[e_i] = 0$, but may possess observation-specific variances $\mathbb{E}[e_i^2] = \sigma_i^2$. In the wage application, x_i collects the high-dimensional worker and firm indicators whose dispersion we ultimately want to measure, while z_i collects the observed covariates (age, firm inputs, and so on) whose effect on wages we want to control for without committing to a functional form.

Our object of interest is a quadratic form $\theta := \beta' A \beta$ for some known non-random symmetric matrix $A \in \mathbb{R}^{p \times p}$ of rank r. The matrix A selects which feature of β to isolate: different choices pick out the variance of worker effects, the variance of firm effects, or their covariance. This quadratic form covers many quantities of economic interest; two examples are summarized next.

Example 1 (Generalized analysis of variance)**.** Consider observations arranged into N groups with T_g observations in the g-th group. Since Fisher (1928), the standard analysis of variance model assumes:

$$y_{gt} = \alpha_g + z_{gt}'\delta + \varepsilon_{gt}, \quad g = 1, \dots, N, \quad t = 1, \dots, T_g,$$

where α_g are group-specific fixed effects, and z_{gt} is a vector of exogenous covariates. The focus here is the variability in the outcome variable attributable to groups, or

$$\sigma_\alpha^2 = \frac{1}{n}\sum_{g=1}^{N} T_g\left(\alpha_g - \bar{\alpha}\right)^2$$

with $n := \sum_{g=1}^{N} T_g$, and $\bar{\alpha} := n^{-1}\sum_{g=1}^{N} T_g\alpha_g$. Our approach relaxes the linearity assumption on covariates and instead assumes any form of unspecified functional dependence, that is

$$y_{gt} = \alpha_g + f(z_{gt}) + \varepsilon_{gt}, \quad f \in \mathcal{F}, \quad g = 1,\dots,N, \quad t = 1,\dots,T_g.$$

We can represent it as in (1), defining $i := i(g,t)$ with $i(\cdot,\cdot)$ being a bijective function, $y_i := y_{gt}$, $z_i := z_{gt}$ and $e_i := \varepsilon_{gt}$,

$$x_i := d_i, \quad \beta := (\alpha_1,\dots,\alpha_N)', \quad d_i := (\mathbb{1}\{g=1\},\dots,\mathbb{1}\{g=N\})'.$$

Our object of interest σ_α^2 can be represented here as $\beta' A\beta$, with

$$A := \begin{pmatrix} A_d'A_d & 0 \\ 0 & 0 \end{pmatrix}, \quad A_d := \frac{1}{\sqrt{n}}(d_1 - \bar{d},\dots,d_n - \bar{d}), \quad \bar{d} := \frac{1}{n}\sum_{i=1}^{n} d_i.$$

Example 2 (Generalized AKM)**.** Our second and leading example is a classic wage decomposition model proposed in Abowd, Kramarz, and Margolis (1999). It models the log wage determination as an additive function of worker fixed effects, firm fixed effects, and a linear function of strictly exogenous covariates. Specifically,

$$y_{gt} = \alpha_g + \psi_{j(g,t)} + z_{gt}'\delta + \varepsilon_{gt}, \quad g = 1,\dots,N, \quad t = 1,\dots,T_g.$$

Here, α_g and $\psi_{j(g,t)}$ capture the gth worker and jth firm unobserved heterogeneity component respectively, and z_{gt} is a vector of exogenous regressors. The bijective function $j(\cdot,\cdot) : \{1,\dots,N\}\times\{1,\dots,\max_g T_g\} \to \{0,\dots,J\}$ maps $n = \sum_{g=1}^{N} T_g$ person-year observations to one of $J+1$ firms. One of the model's objectives is to quantify how much of the variability in log wages is determined by firms,

$$\sigma_\psi^2 = \frac{1}{n}\sum_{g=1}^{N}\sum_{t=1}^{T_g}\left(\psi_{j(g,t)} - \bar{\psi}\right)^2,$$

where $\bar{\psi} = \frac{1}{n}\sum_{g=1}^{N}\sum_{t=1}^{T_g}\psi_{j(g,t)}$. Given evidence that returns to worker characteristics are nonlinear (Mincer 1974; Card et al. 2018), part of the measured variance of worker and firm fixed effects may stem from the linear restriction on z_{gt}. Our methodology replaces it with unspecified functional dependence,

$$y_{gt} = \alpha_g + \psi_{j(g,t)} + f(z_{gt}) + \varepsilon_{gt}, \quad f \in \mathcal{F}, \quad g = 1,\dots,N, \quad t = 1,\dots,T_g.$$

This formulation can capture nonlinear interactions between worker-level and firm-level characteristics that vary across observed groups in the population.

Given that the common covariates z_{gt} and the firm assignments $j(\cdot,\cdot)$ obey a strict exogeneity condition, we can rewrite the equation above as in (1) with

$$x_i := (d_i', h_i')', \quad \beta := (\alpha', \psi')', \quad \alpha := (\alpha_1, \ldots, \alpha_N)' + \mathbb{1}_N' \psi_0, \quad \psi = (\psi_1, \ldots, \psi_J)' - \mathbb{1}_J' \psi_0,$$

defining y_i, z_i, and e_i as in Example 1, and $h_i := (\mathbb{1}\{j(g,t) = 1\}, \ldots, \mathbb{1}\{j(g,t) = J\})'$. The parameter of interest σ_ψ^2 then can be rewritten as $\beta' A_\psi \beta$ with

$$A_\psi := \begin{pmatrix} 0 & 0 & 0 \\ 0 & A_h' A_h & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A_h := \frac{1}{\sqrt{n}}(h_1 - \bar{h}, \ldots, h_n - \bar{h}), \quad \bar{h} := \frac{1}{n}\sum_{i=1}^{n} h_i.$$

2.1. Why existing estimators do not directly apply

Throughout, our object of interest is the quadratic form $\theta = \beta' A \beta$, defined on the high-dimensional fixed-effect coefficients, while the control function $f(z)$ is estimated through a growing approximation space. Several existing estimation strategies address parts of this setting; we review each in turn and explain why none covers the full problem.

KSS-style leave-out estimation. Kline, Saggio, and Sølvsten (2020), hereafter KSS, provide valid leave-out estimation of the same quadratic estimand $\theta = \beta' A \beta$ in finite-dimensional linear many-regressor models under arbitrary heteroskedasticity. Our setting preserves that object, but drops the assumption that observables enter through a finite-dimensional linear control. Once $f(z)$ is replaced by a series approximation $p_k(z)'\alpha$ with $k \to \infty$, validity is no longer automatic: both $\hat{\beta}$ and the leave-out variance estimators inherit approximation error, so one must re-establish that leave-out validity survives series approximation.[1]

Semiparametric series and orthogonalization. Series estimators in partially linear models (Donald and Newey 1994; Cattaneo, Jansson, and Newey 2018a) and orthogonalized or cross-fit procedures (Chernozhukov et al. 2018; Bonhomme, Jochmans, and Weidner 2025) allow an unknown nuisance function, but they estimate low-dimensional slope coefficients or other approximately linear functionals. None provides inference for quadratic forms under the high-dimensional fixed-effect structure and heteroskedasticity that define our setting.

1. A separate strand of the literature relaxes the economic structure of AKM itself. For example, Bonhomme, Lamadon, and Manresa (2019) model latent worker and firm types and reduce dimensionality through grouping. These approaches replace the original variance component with a different earnings structure, rather than preserving $\theta = \beta' A \beta$ under an unknown nuisance function.

3. Finite-sample properties

From now on, we restrict the analysis to model (1). To derive an estimator of β, one must regress y_i on x_i and *functions* of z_i. To this end, let $p^1(z), \ldots, p^k(z)$ be some *approximating* functions, and let $p_k(z) := (p^1(z), \ldots, p^k(z)) \in \mathbb{R}^k$ collect them into a vector. We assume that the class of functions $\mathcal{F}$ to which f belongs can be well approximated by linear combinations of the basis functions $p^1(z), \ldots, p^k(z)$.

Define M_{ij} as a (i,j)th element of $M := I_n - P_k(P_k'P_k)^{-1}P_k'$ with

$$P_k := (p_k(z_1), \ldots, p_k(z_n)) \in \mathbb{R}^{n \times k},$$

and let the partialed-out design matrix $S_{xx} := \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i x_j'$ have a full column rank. Then, the estimator of β is defined as

$$\hat{\beta} := S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i y_j.$$

Under some regularity conditions, the bias of $\hat{\beta}$ is negligible for large k. Rewrite it as:

$$\begin{aligned}
\hat{\beta} &= S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i \left(x_j'\beta + f(z_j) + e_j \right) \\
&= S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i x_j' \beta + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i f(z_j) + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i e_j \qquad (2) \\
&= \beta + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i f(z_j) + S_{xx}^{-1} \sum_{i=1}^n \sum_{j=1}^n M_{ij} x_i e_j := \beta + \mathcal{B} + \mathcal{U}.
\end{aligned}$$

Because of the zero-mean assumption on errors, $\mathbb{E}[\mathcal{U}] = 0$ holds, and the bias in the estimator is reflected in the $\mathcal{B}$ term only. Under some conditions discussed below,

$$\mathcal{B} = o(1)$$

as the number of approximating functions goes to infinity, $k \to \infty$.

Analogously to the fully parametric setup examined by Kline, Saggio, and Sølvsten (2020), the plug-in estimator of the quadratic form $\hat{\theta}_{\mathrm{PI}} := \hat{\beta}' A \hat{\beta}$ is biased, because

$$\mathbb{E}[\hat{\theta}_{\mathrm{PI}} - \theta] = \operatorname{trace}(A \operatorname{var}[\hat{\beta}]) = \sum_{i=1}^n B_{ii} \sigma_i^2,$$

where $B_{ii} := \sum_{j=1}^n M_{ij} x_j' S_{xx}^{-1} A S_{xx}^{-1} \sum_{j=1}^n M_{ij} x_j$ measures the influence of the ith squared error e_i^2 on $\hat{\theta}_{\mathrm{PI}}$. In cases when $p/n \approx 0$, individual elements B_{ii} are close to zero, and the bias is negligible. However, with increasing p, the bias is more pronounced and needs to be accounted for. The intuition is that $\hat{\beta}' A \hat{\beta}$ squares the estimation noise in $\hat{\beta}$ together with the signal, so it overstates the true dispersion $\beta' A \beta$. The leave-out correction estimates that noise term observation by observation and subtracts it, using for unit i a residual variance that never draws on unit i's own outcome.

We introduce some notation to describe our estimator. Collect x_i in a matrix as $X := (x_1, \ldots, x_n)' \in \mathbb{R}^{n \times p}$. Then, $W := (X, P_k) \in \mathbb{R}^{n \times (p+k)}$ is a matrix containing the full set of regressors, with individual vectors w_i, and $W = (w_1, \ldots, w_n)'$. Define

$$\hat{\gamma} := \left(\sum_{j=1}^{n} w_j w_j' \right)^{-1} \sum_{j=1}^{n} w_j y_j,$$

and the leave-one-out version of it,

$$\hat{\gamma}_{-i} := \left(\sum_{j=1}^{n} w_j w_j' - w_i w_i' \right)^{-1} \left(\sum_{j=1}^{n} w_j y_j - w_i y_i \right).$$

Note that the first p elements of $\hat{\gamma}$ are $\hat{\beta}$, and the remaining k elements are basis coefficients from the function approximation. Similarly to Kline, Saggio, and Sølvsten (2020), we estimate θ as

$$\hat{\theta} := \hat{\beta}' A \hat{\beta} - \sum_{i=1}^{n} B_{ii} \hat{\sigma}_i^2, \tag{3}$$

where $\hat{\sigma}_i^2$ is a leave-one-out estimator of the individual variance of e_i defined as

$$\hat{\sigma}_i^2 := y_i \left(y_i - w_i' \hat{\gamma}_{-i} \right). \tag{4}$$

We note that computing $\hat{\gamma}_{-i}$ for each $i = 1, \ldots, n$ is computationally costly in large-scale applications. To avoid this, we represent (4) as

$$\hat{\sigma}_i^2 = \frac{y_i \hat{e}_i}{M_{W,ii}}, \tag{5}$$

where $M_{W,ii}$ is ith diagonal element of the matrix $M_W := I_n - W(W'W)^{-1}W'$ projecting onto the complement of the column space spanned by x_i and functions of z_i, and $\hat{e}_i := \sum_{j=1}^{n} M_{ij}(y_j - x_j \hat{\beta})$ are residuals.

Our estimator is general in the sense that it nests the linear leave-out estimator of KSS as a special case. The generality comes from enlarging the assumed function class from the linear class $\mathcal{F}_{\text{linear}}$ to the broader class $\mathcal{F}$ of nonlinear functions satisfying Assumption 1, so that $\mathcal{F}_{\text{linear}} \subseteq \mathcal{F}$.

To see this more clearly, note that our estimator $\hat{\theta}$ is *exactly* unbiased if the underlying functional class is linear, $f \in \mathcal{F}_{\text{linear}}$. It follows from decomposition as in (2) and noting that the $\mathcal{B}$ term disappears by the properties of the M matrix that projects onto the complement of the space spanned by z_i (where we set $P_k := P_d = (z_1, \ldots, z_n) \in \mathbb{R}^{n \times d}$), in which case $\sum_{i=1}^{n} \sum_{j=1}^{n} M_{ij} x_i z_j' \alpha = 0$. Under the condition that the unknown function $f(z_i) = z_i' \alpha$ is linear in z (as in Kline, Saggio, and Sølvsten (2020)), as a result $\mathbb{E}[\hat{\beta}] = \beta$

holds, and we have

$$
\begin{aligned}
\mathbb{E}[\hat{\sigma}_i^2] &= \mathbb{E}\left[y_i\left(y_i - x_i'\hat{\beta}_{-i} - p_k(z_i)'\hat{\alpha}_{-i}\right)\right] \\
&= \mathbb{E}\left[\left(x_i'\beta + f(z_i) + e_i\right)\left(x_i'\beta + f(z_i) + e_i - x_i'\hat{\beta}_{-i} - p_k(z_i)'\hat{\alpha}_{-i}\right)\right] \\
&= \mathbb{E}\left[\left(x_i'\beta + f(z_i) + e_i\right)\left(x_i'(\beta - \hat{\beta}_{-i}) + (f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}) + e_i\right)\right] \\
&= \left(x_i'\beta x_i' + f(z_i)x_i'\right)\mathbb{E}[\beta - \hat{\beta}_{-i}] + \left(x_i'\beta + f(z_i)\right)\mathbb{E}[f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}] + \sigma_i^2 \\
&= \sigma_i^2
\end{aligned}
$$

where we use the fact that $\mathbb{E}[e_i x_i'(\beta - \hat{\beta}_{-i})] = \mathbb{E}[e_i]\mathbb{E}[x_i'(\beta - \hat{\beta}_{-i})] = 0$, because $\hat{\beta}_{-i}$ is independent of e_i, and similarly $\mathbb{E}[e_i(f(z_i) - p_k(z_i)'\hat{\alpha}_{-i})] = \mathbb{E}[e_i]\mathbb{E}[f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}] = 0$, because $\hat{\alpha}_{-i}$ is independent of e_i.

However, for more general classes of functions $\mathcal{F}$, the exact unbiasedness of $\hat{\beta}$ and $\hat{\sigma}_i^2$ does not hold. Instead, we study an asymptotic sequence in which the number of approximating functions diverges, $k \to \infty$, and show that the bias terms vanish under explicit conditions. Concretely,[2]

$$
\mathbb{E}[\hat{\beta}] - \beta = o(1), \quad \mathbb{E}[\hat{\sigma}_i^2] - \sigma_i^2 = o_p(1), \quad \mathbb{E}[\hat{\theta}] - \theta = o_p(1),
$$

along sequences satisfying the assumptions below.

For these purposes, we assume that the unknown function $f(z)$ belongs to a class $\mathcal{F}$ of smooth functions that can be well approximated by linear combinations of the basis functions $p^1(z), \ldots, p^k(z)$.

Assumption 1 (Function class). *We assume that $f \in \mathcal{F}$, where*

$$
\mathcal{F} := \left\{ f : \min_{\alpha \in \mathbb{R}^k} \mathbb{E}\left[|f(z_i) - p_k(z_i)'\alpha|^2\right] \leq Ck^{-2\alpha_f}, \alpha_f > 1 \right\} \tag{6}
$$

for some absolute constant $C < \infty$. Furthermore, $\sup_{f \in \mathcal{F}} f < M$ for some absolute constant $M < \infty$.

This implies that

$$
\sup_{f \in \mathcal{F}} \mathbb{E}\left[|f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}|^2\right] = \mathcal{O}_p(k^{-2\alpha_f}),
$$

so the mean-square approximation error decays at rate $k^{-2\alpha_f}$ uniformly over $\mathcal{F}$.

The constant α_f governs how fast approximation bias decays. To obtain consistency for the quadratic estimand studied here, we strengthen the usual semiparametric requirement from $\alpha_f > 0$ in Donald and Newey (1994) and Cattaneo, Jansson, and Newey (2018a) to $\alpha_f > 1$: compared with slope-type estimands, quadratic forms require faster decay of approximation error. For polynomial or spline series on compact support with d_{cont} continuous covariates, $\alpha_f = s_f / d_{\text{cont}}$ where s_f is the number of continuous derivatives

2. For expositional purposes, we sometimes drop the Euclidean norm on the vector $\|v\|$, and write asymptotic results as, say, $v = o_p(1)$ instead of $\|v\| = o_p(1)$.

(Chen 2007), so the assumption reduces to $s_f > d_{\text{cont}}$: the unknown function must have more continuous derivatives than the number of continuous covariates entering flexibly. Discrete covariates such as gender, occupation, or region do not count toward d_{cont} because they are absorbed by group interactions rather than by the polynomial basis. In the wage applications of Section 8, the parsimonious sample has $d_{\text{cont}} = 1$ (age), so $s_f > 1$ suffices—the wage–age profile need only be twice continuously differentiable. The Panel B specification adds employment, fixed assets, and intermediate inputs, giving $d_{\text{cont}} \leq 4$ and requiring $s_f > 4$. Both conditions are mild for smooth economic relationships. The condition is sufficient but not shown to be necessary. Even though the assumption on α_f is strengthened compared to the standard linear setups, it is equivalent to the assumption commonly placed in settings with nonlinear structure, as in, for example, Hirano, Imbens, and Ridder (2003) and Farrell, Liang, and Misra (2021).

Remark 1. *The results cover fixed-dimensional continuous covariates and fixed collections of discrete interaction groups, as in Section 8. They do not cover growing continuous dimension, growing numbers of interaction groups, or machine-learning nuisance classes.*

4. Consistency

In this section, we prove the consistency result for the proposed estimator $\hat{\theta}$. We study the asymptotic behavior of $\hat{\theta}$ assuming that x_i, z_i, and A are sequences of constants so that the only source of randomness is e_i. We adopt the conditional perspective of Scheffé (1959), Searle, Casella, and McCulloch (2009), and Kline, Saggio, and Sølvsten (2020), treating x_i, z_i, and A as fixed. This allows us to be agnostic about potential dependence between x_i, z_i, and A. Our analysis differs from Cattaneo, Jansson, and Newey (2018b), who consider sequences of random variables and condition on z_i only. Limits are taken assuming the number of observations goes to infinity, $n \to \infty$, the number of approximating functions goes to infinity, $k \to \infty$ (so that the finite-sample bias of $\hat{\beta}$ and $\hat{\sigma}_i^2$ is asymptotically of negligible order), and the dimensionality of x_i goes to infinity, $p \to \infty$, to model the limited mobility bias. We make the following additional assumptions:

Assumption 2 (Data-generating process)**.** *(i)* $\max_i(\mathbb{E}[e_i^4] + \sigma_i^{-2}) = \mathcal{O}(1)$; *(ii) there exists a* $c < 1$ *such that* $\max_i P_{W,ii} < c$ *for all* n*, where* $P_{W,ii} := 1 - M_{W,ii}$*; (iii)* $\max_i(x_i'\beta)^2 = \mathcal{O}(1)$.

Part *(i)* excludes heavy-tailed distributions of the errors as is typically assumed in the literature on OLS estimation (Cattaneo, Jansson, and Newey 2018b; Kline, Saggio, and Sølvsten 2020). Parts *(ii)* and *(iii)* coupled with the Assumption 1 imply that the leave-one-out estimator $\hat{\sigma}_i^2$ is well-defined and has bounded variance. Part *(ii)* also implies that $\frac{p+k}{n} \leq c < 1$ for all n.

Assumption 3 (Growth rate conditions)**.** *We assume that* $k \to \infty$ *and* $p \to \infty$ *as* $n \to \infty$*, and* $k = o(n)$, $k = o(p)$, $n = o(k^{\alpha_f})$, *and* $p = \mathcal{O}(n)$.

This assumption places restrictions on the relationship between growth rates of the number of observations, linear covariates, and approximating functions. In particular, it reflects empirical practice in the AKM literature where the number of observations is of a much larger magnitude than the potential number of series terms. It also places a lower bound on the number of series terms, since k must be large enough that the n/k^{α_f} term is small. Intuitively, the window $n^{1/\alpha_f} \ll k \ll n$ balances zero asymptotic bias against a negligible added variance contribution. Relaxing the assumption to cover $k = \mathcal{O}(n)$ would be of independent theoretical interest, though of limited empirical relevance in the AKM setting. We conjecture that letting the number of series terms grow proportionally with the number of observations would add a further term to the asymptotic variance of $\hat{\theta}$, in line with the theory on linear functionals with growing dimension (Cattaneo, Jansson, and Newey 2018b). We leave a proof to future work.

Shared rate restrictions for the proved results. The results below all rely on the same core asymptotics: the series dimension grows ($k \to \infty$), total regressor dimension remains bounded away from the sample size through $\max_i P_{W,ii} < c < 1$, and approximation bias decays fast enough through $\alpha_f > 1$. The theorem-specific conditions then add the restrictions needed for each result: Lemma 1 requires $\operatorname{trace}(\tilde{A}^2) = o(1)$, Theorem 3 adds the fixed-rank leverage condition $\max_i v_i' v_i = o(1)$, and Theorem 4 adds the growing-rank negligibility conditions on $\max_i \left((\tilde{w}_i'\gamma)^2 + (\check{w}_i'\gamma)^2\right)$ and on the leading eigenvalue share $\lambda_1^2 / \sum_{\ell=1}^{r} \lambda_\ell^2$. The tradeoff behind the fixed-dimensional series results is simple: k must grow fast enough for approximation error to vanish, but not so fast that the combined regressor dimension $p + k$ drives leverage close to one.

Relative to linear leave-out estimation, replacing linear controls by a growing basis adds one additional task: one must show that the approximation terms from the unknown nuisance are negligible both for $\hat{\beta}$ and for the leave-out residual-variance correction. Lemma 1 and Theorems 3–4 establish exactly this under Assumptions 1–3.

Corollary 1. *Suppose the covariates entering the unknown function can be written as $z_i = (u_i', s_i)'$, where $u_i \in \mathbb{R}^{d_{\text{cont}}}$ has fixed dimension, $s_i \in \{1, \ldots, G\}$ indexes a fixed collection of discrete groups, and*

$$f(z_i) = \sum_{g=1}^{G} \mathbb{1}\{s_i = g\} f_g(u_i).$$

Assume the support of u_i is compact and that, for the specified basis $\{p^j(u_i)\}_{j=1}^{k}$, each f_g satisfies

$$\inf_{\alpha_g \in \mathbb{R}^k} \mathbb{E}\left[\left|f_g(u_i) - p_k(u_i)'\alpha_g\right|^2\right] \le C k^{-2 s_f / d_{\text{cont}}}$$

for a common constant $C < \infty$ and common smoothness index $s_f > d_{\text{cont}}$, uniformly over g. Then the interacted basis $\{\mathbb{1}\{s_i = g\} p^j(u_i)\}_{g,j}$ satisfies Assumption 1 with $\alpha_f = s_f / d_{\text{cont}} > 1$. If, in addition, Assumptions 2 and 3 hold together with the theorem-specific leverage and eigenvalue conditions, then the conclusions of Lemma 1, Theorem 3, and Theorem 4 apply to this interacted specification.

Let $\tilde{A} := S_{xx}^{-1/2} A S_{xx}^{-1/2}$ and write its eigendecomposition as $\tilde{A} = Q \Lambda Q'$, where $\Lambda =$

$\mathrm{diag}(\lambda_1, \ldots, \lambda_r)$ collects the r nonzero eigenvalues and $Q \in \mathbb{R}^{p \times r}$ the corresponding eigenvectors.

Lemma 1. *If Assumptions 1, 2, and 3 hold, A is positive semi-definite, $\theta = \beta' A \beta = \mathcal{O}(1)$, and trace$(\tilde{A}^2) = \sum_{\ell=1}^{r} \lambda_\ell^2 = o(1)$, then*

$$\hat{\theta} - \theta \xrightarrow{p} 0.$$

5. Large-scale approximation

In this section we discuss an alternative estimator that allows for fast computation in typical large-scale applications such as those based on administrative linked employer-employee data. We follow Kline, Saggio, and Sølvsten (2020) by considering the random projection method of Achlioptas (2003), known as the Johnson–Lindenstrauss approximation (JLA), which names the estimator below. To describe it, fix $m \in \mathbb{N}$ and generate matrices $R_B, R_P \in \mathbb{R}^{m \times n}$ where each (i, j) coordinate is a random draw from the following Rademacher distribution:

$$R_{\cdot,ij} = \begin{cases} +1 & \text{with probability } 1/2, \\ -1 & \text{with probability } 1/2. \end{cases}$$

Decompose $A = 1/2(A_1' A_2 + A_2' A_1)$ for $A_1, A_2 \in \mathbb{R}^{n \times p}$, where $A_1 = A_2$ if A is positive semi-definite. Denote

$$\hat{P}_{W,ii} := \frac{1}{m} \left\| R_P W S_{ww}^{-1} w_i \right\|^2, \quad \hat{B}_{ii} := \frac{1}{m} \left(R_B A_1 S_{xx}^{-1} \sum_{j=1}^{n} M_{ij} x_j \right)' \left(R_B A_2 S_{xx}^{-1} \sum_{j=1}^{n} M_{ij} x_j \right).$$

The proposed estimator is then:

$$\hat{\theta}_{\text{JLA}} := \hat{\beta}' A \hat{\beta} - \sum_{i=1}^{n} \hat{B}_{ii} \hat{\sigma}_{i,\text{JLA}}^2, \quad \hat{\sigma}_{i,\text{JLA}}^2 := \frac{y_i (y_i - w_i' \hat{\gamma})}{1 - \hat{P}_{W,ii}} \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right). \tag{7}$$

As in Kline, Saggio, and Sølvsten (2020), the term $\frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}}$ removes a non-linearity bias from the approximation of $P_{W,ii}$ by $\hat{P}_{W,ii}$.

Lemma 2. *If Assumptions 1, 2, and 3 are satisfied, $n/m^4 = o(1)$, $\mathrm{var}[\hat{\theta}]^{-1} = \mathcal{O}(n)$, and one of the following conditions hold, then $\mathrm{var}[\hat{\theta}]^{-1/2}(\hat{\theta}_{JLA} - \hat{\theta} - \mathrm{B}_m) = o_p(1)$ where $|\mathrm{B}_m| \leq \frac{1}{m} \sum_{i=1}^{n} P_{W,ii}^2 |B_{ii}| \sigma_i^2$:*

(i) A is positive semi-definite and $\mathbb{E}[\hat{\beta}' A \hat{\beta}] - \theta = \sum_{i=1}^{n} B_{ii} \sigma_i^2 = \mathcal{O}(1)$.

(ii) $A = 1/2(A_1' A_2 + A_2' A_1)$ where $\theta_1 = \beta' A_1' A_1 \beta$ and $\theta_2 = \beta' A_2' A_2 \beta$ satisfy (i) and $\frac{\mathrm{var}[\hat{\theta}_1]\, \mathrm{var}[\hat{\theta}_2]}{n\, \mathrm{var}[\hat{\theta}]^2} = \mathcal{O}(1)$.

6. Limiting distributions

This section derives two limiting distribution results for $\hat{\theta}$. The first assumes that the rank r of A is fixed; the second allows r to grow with n.

6.1. Fixed rank

First, we derive the limiting distribution of $\hat{\theta}$ for fixed r. Using the eigendecomposition $\tilde{A} = Q\Lambda Q'$ introduced before Lemma 1, we represent the estimator as

$$\hat{\theta} = \sum_{\ell=1}^{r} \lambda_\ell \left(\hat{b}_\ell^2 - \widehat{\operatorname{var}}[\hat{b}_\ell] \right),$$

where $\hat{b} := \sum_{i=1}^n v_i y_i$, $\widehat{\operatorname{var}}[\hat{b}] := \sum_{i=1}^n v_i v_i' \hat{\sigma}_i^2$, and $v_i := Q' S_{xx}^{-1/2} \sum_{j=1}^n M_{ij} x_j$.

The next theorem gives the fixed-rank limit for $\hat{\theta}$ and consistency of $\widehat{\operatorname{var}}[\hat{b}]$.

Theorem 3. *If Assumptions 1, 2, and 3 hold, r is fixed, and $\max_i v_i' v_i = o(1)$, then*

(i) $\operatorname{var}[\hat{b}]^{-1/2}(\hat{b} - b) \xrightarrow{d} \mathcal{N}(0, I_r)$, *where* $b := Q' S_{xx}^{1/2} \beta$,

(ii) $\operatorname{var}[\hat{b}]^{-1} \widehat{\operatorname{var}}[\hat{b}] \xrightarrow{p} I_r$,

(iii) $\hat{\theta} = \sum_{\ell=1}^r \lambda_\ell \left(\hat{b}_\ell^2 - \operatorname{var}[\hat{b}_\ell] \right) + o_p(\operatorname{var}[\hat{\theta}]^{1/2})$.

The growing-rank case is more relevant for AKM-type applications, so we turn to that setting next.

6.2. Growing rank

Define $\tilde{w}_i := \breve{A} S_{ww}^{-1} w_i$, where $\breve{A} := \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}$ and $\gamma := \begin{pmatrix} \beta & \alpha \end{pmatrix}'$. Then

$$\theta = \gamma' \breve{A} \gamma = \gamma' S_{ww} S_{ww}^{-1} \breve{A} \gamma = \sum_{i=1}^n \gamma' w_i \tilde{w}_i' \gamma$$

and the leave-one-out estimator can be written as

$$\hat{\theta} = \sum_{i=1}^n y_i \tilde{w}_i' \hat{\gamma}_{-i},$$

where the approximation error from replacing $\gamma' w_i$ by y_i is controlled by Assumption 1 and 3. A direct algebraic rearrangement yields

$$\hat{\theta} = \sum_{i=1}^n \sum_{\ell \neq i} C_{i\ell} y_i y_\ell,$$

where $C_{i\ell} := B_{W,i\ell} - 2^{-1} M_{W,i\ell} \left(B_{W,ii} M_{W,ii}^{-1} + B_{W,\ell\ell} M_{W,\ell\ell}^{-1} \right)$. Thus $\hat{\theta}$ is a second-order U-statistic with sample-dependent kernel $C_{i\ell}$. Theorem 4 combines this representation with

Appendix Lemma 3; relative to Kline, Saggio, and Sølvsten (2020), the only additional term is the series approximation error, which is negligible under Assumption 1 and 3.

Define also

$$\check{w}_i := \sum_{\ell=1}^{n} M_{W,i\ell} \frac{B_{W,\ell\ell}}{1 - P_{W,\ell\ell}} w_\ell.$$

Theorem 4. *If Assumption 1, 2, and 3 hold, and the following conditions are satisfied,*

$$\textit{(i)}\ \operatorname{var}[\hat{\theta}]^{-1} \max_i \left((\tilde{w}_i'\gamma)^2 + (\check{w}_i'\gamma)^2 \right) = o(1), \quad \textit{(ii)}\ \frac{\lambda_1^2}{\sum_{\ell=1}^{r} \lambda_\ell^2} = o(1),$$

then $\operatorname{var}[\hat{\theta}]^{-1/2}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, 1)$.

7. Simulation study

We use a Monte Carlo study to evaluate the estimator under controlled conditions that isolate the problems described in the theory. In the baseline design, the nuisance covariates z_i are drawn uniformly on a fixed cube, the linear regressors x_i are nonlinear functions of z_i, and the quadratic form of interest is $\theta = \beta' A \beta$, where A selects the first two coordinates of β, so it measures dispersion in two fixed effects. Unless otherwise noted, the nonlinear designs use $n = 500$, nuisance dimension $d = 4$, and $p = 90$ linear regressors. The rows are constructed as comparative statics: within a panel, the simulation holds the underlying random design fixed whenever dimensions permit and changes only the feature named in the row. We vary the nonlinearity of $f(z)$, the error distribution, the dimension of the nuisance, the number of linear regressors, leverage, and sample size.

More specifically, each scenario first draws a fixed design $(z_i, x_i)_{i=1}^n$ and coefficient vector β, then holds that design fixed across Monte Carlo replications while redrawing only the error term e_i. The first panel of Table 1 uses the same z_i, x_i, β, and error draws and changes only the functional form of f. The second panel keeps the strongly nonlinear radial design as the reference case; the heteroskedastic and heavy-tailed rows change only the error process, while the remaining rows change the named dimension, regressor count, leverage rule, or sample size using nested random draws where possible. The nuisance covariates satisfy $z_i \sim \text{Unif}([-1, 1]^d)$. Before standardization, the linear regressors are

$$x_{ij}^* = \exp\{0.30 \, \|z_i\|_2 + 0.50\eta_{ij} + 0.10\nu_j\}, \qquad \eta_{ij}, \nu_j \sim \mathcal{N}(0, 1),$$

and each column of x_i^* is centered and scaled. The nuisance signal is also centered and scaled before entering the outcome. We use three functional forms:

$$f_{\text{linear}}(z_i) = 0.8z_{i1} - 0.5z_{i2} + 0.35z_{i3} - 0.2z_{i4}, \quad f_{\text{cubic}}(z_i) = 0.8z_{i1}^3 - 0.5z_{i2}^2 + 0.3z_{i3}z_{i4},$$

and the strongly nonlinear radial design

$$f_{\text{radial}}(z_i) = \|z_i\|_2^7.$$

The radial seventh-power function is the main nonlinear stress test: it differs sharply from any linear control because its slope rises quickly as observations move away from the center of the cube. The error term is standard Gaussian, heteroskedastic Gaussian with scale $0.5 + 0.5|x_{i1}|$ normalized by its fixed-design root mean square, or a variance-one Student-t distribution with five degrees of freedom. These are population or fixed-design normalizations; we do not divide each Monte Carlo draw by its realized sample standard deviation. Most scenarios use $n = 500$ observations per repetition, with a larger-sample design at $n = 1000$; the regressor dimension is either $p = 90$ or $p = 300$, and the nuisance dimension is either $d = 4$ or $d = 8$.[3]

We compare the plug-in estimator with leave-out estimators that use complete total-degree polynomial bases of degree 1, 3, and 5, including all monomials whose total degree does not exceed the stated degree. Cubic polynomials are common in applied work (Card et al. 2018); the degree-5 basis lets us check whether going beyond cubic matters. Table 1 reports the subset of designs that cover the main cases in the paper: approximation error and nonlinearity, robustness to heteroskedasticity and heavy tails, and finite-sample performance as nuisance dimension, the number of regressors, leverage, and sample size change one at a time. The first four columns report signed Monte Carlo bias for the plug-in (PI) and leave-out (LOO) estimators at each basis degree. The last two report RMSE and nominal 95% coverage for LOO at degree 5, with confidence intervals based on the estimated variance of the quadratic form computed from the leave-one-out variances $\hat{\sigma}_i^2$, so the table separates point-estimation performance from finite-sample inference.

Approximation error and the smoothness condition. Assumption 1 predicts that series approximation error decays at rate $k^{-2\alpha_f}$, so higher-degree bases should reduce bias most when f departs strongly from linearity. The simulations confirm this. In the strong radial design, signed bias falls in magnitude from about -0.04 for PI(1) and LOO(1) to -0.02 for degree 3 and -0.01 for degree 5. Under high leverage, the same pattern holds: signed bias falls from about -0.02 for PI(1) and -0.03 for LOO(1) to zero after adding higher-degree basis terms. By contrast, in the linear benchmark and the mildly nonlinear cubic design, signed bias is already close to zero, with the largest entry about 0.01. The first panel of Table 1 should therefore be read as evidence on where the smoothness condition matters in finite samples: adding basis terms removes systematic misspecification bias when nonlinearity is large, and costs little when it is not.

Heteroskedasticity and error tails. The leave-out correction in (3) uses observation-specific residual variances $\hat{\sigma}_i^2$, so it depends on the moment conditions in Assumption 2*(i)*. The heteroskedastic and heavy-tailed designs show why the table reports both signed bias and RMSE. Because these rows now keep the same radial design as the homoskedastic strong-nonlinearity row, the low-degree signed bias is similar: about -0.03 to -0.04

3. In Table 1, $p = 90$ in all scenarios except the many-regressor design, where $p = 300$. The nuisance dimension is $d = 4$ in all scenarios except the higher-dimension design, where $d = 8$. The sample size is $n = 500$ in all scenarios except the larger-sample design, where $n = 1000$.

Table 1: Simulation study: bias and finite-sample performance

Scenario	PI(1)	LOO(1)	LOO(3)	LOO(5)	RMSE(5)	Cov.(5)
Approximation and nonlinearity						
Mild nonlinearity	0.01	0.00	0.00	0.00	0.05	0.93
Linear benchmark	0.00	0.00	0.00	0.00	0.05	0.93
Strong nonlinearity	-0.04	-0.04	-0.02	-0.01	0.05	0.90
Strong radial design: one-feature changes						
Heteroskedasticity	-0.03	-0.04	-0.02	-0.01	0.05	0.90
Heavy-tailed errors	-0.03	-0.04	-0.01	-0.01	0.05	0.91
Higher nuisance dimension	-0.05	-0.05	-0.02	NA	NA	NA
Many regressors	-0.01	-0.03	-0.02	-0.01	0.06	0.97
High leverage	-0.02	-0.03	-0.01	0.00	0.02	0.86
Larger sample	0.02	0.01	-0.01	0.00	0.03	0.95

Note: Columns PI(1), LOO(1), LOO(3), and LOO(5) report signed Monte Carlo bias, $\mathbb{E}[\hat{\theta} - \theta]$, for the plug-in estimator with a degree-1 basis and the leave-out estimator with degree-1, degree-3, and degree-5 bases, respectively. Positive entries indicate overestimation of the quadratic estimand. RMSE(5) and Cov.(5) report the RMSE and nominal 95% coverage of the degree-5 leave-out estimator, with intervals based on the estimated variance of the quadratic form computed from the leave-one-out variances $\hat{\sigma}_i^2$. The first panel varies only the functional form of $f(z)$. The second panel maintains the strong radial design and changes only the feature named in the row, relative to the strong-nonlinearity row. Entries are based on 5,000 Monte Carlo replications for each estimable design. "NA" means that the complete basis is not estimable: with $d = 8$, the degree-5 basis has 1,287 columns, so $p + k > n$ in the higher-dimension row.

before the higher-degree basis removes most of it. Degree-5 RMSE is about 0.05 under both heteroskedasticity and Student-t_5 errors. These designs therefore create dispersion that is not captured by signed bias alone; adding basis terms removes the approximation component, but residual-variance estimation remains a finite-sample issue.

Leverage and many regressors. Assumption 2*(ii)* requires $\max_i P_{W,ii} < c < 1$, bounding individual leverage away from one as the combined regressor dimension $p + k$ grows. The many-regressor and high-leverage designs make this condition hard to satisfy. Point estimation remains reliable: in the many-regressor design, the degree-5 leave-out estimator has signed bias around -0.01 and RMSE of about 0.06. Finite-sample inference is less robust than point estimation. Nominal 95% coverage is about 0.86 in the high-leverage design, 0.97 in the many-regressor design, 0.90 under heteroskedasticity, and 0.91 under Student-t_5 errors. In the higher-dimension design, degree 3 remains feasible and has -0.02 bias and about 0.91 coverage, but the complete degree-5 basis is infeasible and is reported as "NA." The pattern points to residual-variance estimation and leverage, rather than signed point-estimate bias alone, as the main finite-sample problem. Larger samples improve point accuracy: signed bias is near zero, RMSE falls to about 0.03, and coverage returns to about 0.95.

Summary. For applied work, adding basis terms is most valuable when nonlinear misspecification is large, especially under high leverage. When those features are absent, additional flexibility costs little but gains little. The point estimator is reliable across most designs; interval estimates are more fragile and warrant caution when leverage, heteroskedasticity, or nuisance dimension make residual-variance estimates unstable.

AKM-calibrated Monte Carlo. We adapt the calibration logic in Section 9.5 of Kline, Saggio, and Sølvsten (2020) to our application; we do not reproduce their first-differenced mover-pair design. We hold fixed a small connected mover graph with 13,939 person-year observations, 2,393 workers, 639 firms, and 6,366 worker–firm matches, using only gender and age as controls. The OLS worker and firm effects on this graph are rescaled so that their variances match the leave-out estimates. Each match shock is drawn from a standardized Student-t distribution with five degrees of freedom, with a heteroskedastic variance. Following Kline, Saggio, and Sølvsten (2020), we let that variance depend on the firm-variance diagonal B_{gg}, the leverage P_{gg}, and the log employment of the worker's current and other firms through an exponential function fit to the leave-out variance estimates.[4] The design crosses linear and degree-5 nuisance DGPs with linear and degree-5 fitted nuisance specifications, holding firm assignments, covariates, and year assignments fixed across 1,000 draws.

Table 2: Portuguese-graph calibrated Monte Carlo for firm-effect variance

DGP	Nuisance fit	KSS bias (MC SD)	HO bias	PI bias
Linear DGP	Flexible	0.06% (0.50%)	1.24%	1.90%
Linear DGP	Linear	0.06% (0.48%)	1.25%	1.90%
Nonlinear DGP	Flexible	0.07% (0.62%)	1.31%	1.98%
Nonlinear DGP	Linear	0.18% (0.62%)	1.60%	2.37%

Note: The true value is the calibrated firm-effect variance. "KSS bias (MC SD)" reports point-estimate bias relative to that true value, with the Monte Carlo standard deviation of the relative estimation error in parentheses. "HO bias" and "PI bias" report the same relative bias for the homoskedasticity-only correction and the plug-in estimator. "Flexible" denotes the degree-5 nuisance fit and "Linear" the linear fit. Each design uses 1,000 draws.

Table 2 evaluates the point correction. KSS relative bias is 0.06–0.18% across the four designs, compared with 1.24–1.60% for the homoskedastic correction and 1.90–2.37% for the plug-in estimator. The KSS Monte Carlo standard deviation is 0.48–0.62% of the true value. The KSS bias is largest in the misspecified nonlinear-DGP/linear-fit design but stays tiny throughout, because this parsimonious calibration carries only an age profile: the exercise checks that the leave-out point correction stays nearly unbiased on a realistic mobility graph, and does not speak to the functional-form sensitivity in the application, which operates through the firm inputs that this design excludes. The feasible standard-error calculation is not validated by this exercise: it excludes about half of the match-level leave-out variance estimates, produces standard errors roughly six to eight times the Monte Carlo standard deviation, and the nominal 95% intervals cover in all

4. The exponential model is fit by nonlinear least squares in levels to all finite leave-out variance estimates, using the positive estimates only to initialize. Because the KSS regression premultiplies each match mean by the square root of its person-year count, the fitted variance is in weighted-match units; we divide it by the match count before assigning a common shock to the match's person-year observations.

1,000 draws. We treat those interval results as diagnostics, not as evidence for cluster-level inference.

8. Empirical application

We use Portuguese linked employer–employee data to study how alternative control specifications change the worker–firm wage decomposition. Adding worker and firm-input controls lowers all three variance components. Allowing those controls to enter nonlinearly produces smaller movements in firm variance and sorting, while the worker component is more sensitive to the basis used for group-specific controls.

8.1. Data and sample selection

Our empirical analysis combines two Portuguese sources of administrative data. The matched employer–employee dataset *Quadros de Pessoal* (QP) reports worker characteristics (age, gender, occupation, qualification, education, contract type, hours, compensation) and firm characteristics (location, industry, employment) for the universe of private-sector firms. The firm accounts dataset *Central de Balanços* (CB) provides annual balance-sheet and income-statement information, including fixed assets and intermediate inputs, for non-financial corporations. We link QP and CB at the firm level to obtain a panel with worker characteristics and firm inputs.

We restrict our sample to the period 2008–2018 and to full-time workers aged 20–65. The dependent variable is the log of individual hourly wages. Following the standard AKM literature, we decompose the cross-sectional variance of log wages into the variance of worker fixed effects (σ^2_α), the variance of firm fixed effects (σ^2_ψ), and their covariance $\mathrm{cov}(\alpha, \psi)$, which captures sorting. Detailed sample descriptives, estimation-variable coverage, and sample-flow statistics are reported in Appendix Tables B1, B2, and B3.

8.2. Estimation

We study two versions of this empirical exercise. The first is a parsimonious specification that uses only worker-side controls: gender and age. The second specification adds worker education and qualification together with log firm employment, $\log(1 + \text{fixed assets})$, and $\log(1 + \text{intermediate inputs})$. About 7.6% of firm-years report exactly zero fixed assets, so the two monetary inputs enter as $\log(1 + x)$ rather than $\log(x)$ to retain these economically meaningful zeros.[5] The transformations also keep the raw monetary scales from dominating the polynomial basis, and negative accounting entries are treated as missing. In the empirical tables, we label these two exercises as Panel A (parsimonious

5. The transformation $\log(1 + x)$ is not invariant to the units in which x is measured (Chen and Roth 2024): rescaling the monetary inputs changes where the accounting zeros sit relative to the positive values. This bears on the linear specifications more than on the flexible ones, which approximate an arbitrary smooth function of each input and are therefore nearly unaffected by the choice of units. Fixed assets and intermediate inputs are held at their recorded scale throughout, and both enter only as nuisance controls whose coefficients we do not interpret.

controls) and Panel B (worker and firm-input controls). Because Panel B requires non-missing firm controls, the main empirical tables estimate both panels on the Panel B non-missing *leave-match-out* set: the largest connected worker–firm set, on which the KSS bias correction leaves out one entire worker–firm match (all of its person-year observations) at a time rather than a single observation.[6] This keeps the main cross-panel comparison about controls rather than sample composition; Appendix Table B3 reports the larger parsimonious candidate sample as well as the common main estimation sample.

Within each of these two control sets, we estimate the same nested five-step specification ladder, progressively relaxing the restrictions on how covariates enter the wage equation: (i) a pure AKM baseline with no observed controls beyond the fixed effects; (ii) linear additive controls, meaning the standard cubic age profile normalized at 50 together with education, qualification, and firm inputs entering linearly and additively; (iii) heterogeneous linear controls, which add demographic-group deviations in the age profile and group-specific slopes on the firm inputs; (iv) nonlinear homogeneous controls, which replace those terms with a common degree-5 polynomial basis; and (v) the full model, which interacts that basis with demographic groups. The ladder separates the roles of adding observables, allowing slope heterogeneity, and allowing nonlinearity. Because age already enters as a cubic profile in the linear additive rung, the move to the degree-5 basis mainly adds curvature to the firm inputs. The polynomial basis is interacted with a fully saturated set of worker categorical variables, so the full model allows group-specific nonlinear effects of each continuous covariate: in Panel A the interaction groups are defined by gender, while in Panel B they are defined by gender $\times$ education $\times$ qualification.

We use the polynomial basis for the main ladder and re-estimate its two nonlinear rungs with two alternatives. Probabilists' Hermite polynomials span the same degree-5 space as the monomials, so they produce the same fitted wage. They can still shift the decomposition slightly, because the two families place their constant and low-order terms differently relative to the fixed-effect normalization, and that changes how the fitted variation is split between the control profile and the worker and firm effects. Additive cubic B-splines provide a different functional form with quantile-spaced knots. We choose the spline counts so that each alternative has the same realized control dimension as the corresponding polynomial model: $k = 4$ and 9 in Panel A, and $k = 129$ and 3,004 in Panel B for the common and group-specific nonlinear specifications. Appendix Table B6 compares the resulting decompositions.

Mapping to theory. To map the empirical specification into the model in Section 2, let z_i^{cont} denote the continuous covariates that enter flexibly and let $s_i \in \{1, \dots, G\}$ denote the worker categorical group that defines the interaction cell. In Panel A, z_i^{cont} contains age

6. The source panel spans 2008–2018, but fixed assets are unavailable in 2008–2009. After imposing the Panel B non-missing firm-control requirement, the common Panel A–Panel B estimation sample is therefore effectively drawn from 2010–2018; Appendix Table B4 reports the year-by-year joint availability of the firm controls.

and s_i is gender. In Panel B, z_i^{cont} contains age, log employment, $\log(1 + \text{fixed assets})$, and $\log(1 + \text{intermediate inputs})$, while s_i indexes the fixed collection of gender $\times$ education $\times$ qualification cells. Writing x_i for the regressors that remain outside the flexible nuisance term—the worker and firm fixed effects, year effects, and any controls that enter linearly at a given specification—the full specification can be written as

$$y_i = x_i'\beta + f_{s_i}(z_i^{\text{cont}}) + e_i = x_i'\beta + \sum_{s=1}^{G} \mathbb{1}\{s_i = s\}\, f_s(z_i^{\text{cont}}) + e_i.$$

This is a special case of (1): the unknown function is simply evaluated on the enlarged argument $(z_i^{\text{cont}\prime}, s_i)'$, so the nuisance is allowed to be a different smooth function in each worker group.

Our implementation first replaces age with age minus 50, centers the remaining continuous inputs, scales all continuous inputs by their sample standard deviations, approximates each group-specific function f_s with the same polynomial family, and lets the coefficients vary across groups. Concretely, if $\{p^j(z_i^{\text{cont}})\}_{j=1}^k$ is the polynomial basis, then the interacted series uses the terms $\{\mathbb{1}\{s_i = s\}p^j(z_i^{\text{cont}})\}_{s,j}$. Equivalently, the model fits a separate smooth profile of age in Panel A, and separate smooth profiles of age and the three transformed firm inputs in Panel B, within each worker category.

For the theory, these interactions do not increase the continuous dimension of the approximation problem. The indicators $\mathbb{1}\{s_i = s\}$ split the sample into a fixed number of groups, but smoothness is still imposed only with respect to the continuous arguments in z_i^{cont}. Therefore Assumption 1 applies within each group at rate $\alpha_f = s_f/d_{\text{cont}}$, where d_{cont} is the number of continuous covariates entering the nuisance and s_f is the number of their continuous derivatives. This is precisely the fixed-dimensional interacted-series case covered by Corollary 1: $d_{\text{cont}} = 1$ in Panel A and $d_{\text{cont}} = 4$ in Panel B.

There are two distinctions between the formal estimator and the empirical correction. First, the theory is stated for independent observation-level errors and deletes one observation at a time, whereas the application collapses person-years to weighted worker–firm matches and deletes an entire match. Second, the application estimates controls and year effects jointly with worker and firm effects, forms a residualized wage, and then applies the fixed-effect-only KSS correction. Frisch–Waugh–Lovell equivalence preserves the plug-in fixed-effect components but not the augmented leverage and leave-out quantities used by the estimator in Section 3. The empirical exercise is therefore a specification diagnostic based on residualized leave-match-out KSS point estimates, not a direct implementation of the augmented-regressor estimator proved above. A formal cluster-level extension and a full augmented implementation would require additional work.

The interaction groups are defined from worker-side categorical variables, not firm identifiers. This matters for identification. Interacting the series with firm identifiers would produce columns that are nearly collinear with the firm fixed effects already contained in X, undermining the full-rank condition on $S_{xx} = X'MX$ required by Assumption 2.[7]

7. Age is normalized at 50 throughout, and the common linear age term is excluded because it is not

The firm-side controls that distinguish Panel B from Panel A are employment, fixed assets, and intermediate inputs. Appendix Table B4 reports their within-firm variation and missingness in the cleaned firm-year panel. On their estimation scales, their within-firm variance shares are about 7.9% for employment, 17.8% for fixed assets, and 9.8% for intermediate inputs, so most of their variation is cross-sectional. On the raw headcount and euro scales the corresponding within-firm shares are only 5.8%, 1.6%, and 5.0%: differences in firm size across firms dominate the year-to-year movements within a firm. Taking logs removes that size gap and raises the within-firm share of fixed assets more than tenfold, from 1.6% to 17.8%. Specification diagnostics for each specification are reported in Appendix Table B7.

8.3. Results

Table 3 reports the bias-corrected variance components for each specification on the common Panel B leave-match-out sample. We report point estimates and do not attach standard errors because the application uses a cluster-level leave-match-out correction outside the observation-level theory developed above. Cross-specification differences are descriptive. Table 4 reports $\mathrm{var}(\tilde{y}) / \mathrm{var}(y)$, the ratio of residualized-wage variance to total wage variance after partialling out observables and year effects; values above one signal that near-collinear controls are amplifying rather than absorbing variance. Appendix Table B5 decomposes the within-panel movements into stepwise changes along two paths—one adding heterogeneity first, the other nonlinearity first—both arriving at the full model. Appendix Table B6 compares the nonlinear estimates across polynomial, Hermite, and cubic B-spline bases at the same realized control dimension.

Panel A: parsimonious controls. Panel A is stable across the specification ladder. Worker variance ranges from 0.534 to 0.563, firm variance from 0.129 to 0.136, and sorting from 0.082 to 0.088. The residualized-wage variance ratio ranges from 0.956 to 0.998. Thus neither the standard age profile nor its heterogeneous and nonlinear extensions materially changes the decomposition on the common Panel B sample.

Panel B: worker and firm-input controls. Adding education, qualification, and the transformed firm inputs changes the decomposition already in the linear additive specification. Relative to the AKM baseline, the additive specification lowers worker variance from 0.553 to 0.478, firm variance from 0.136 to 0.115, and sorting from 0.088 to 0.053. Allowing heterogeneous linear slopes lowers worker variance further to 0.455, while firm variance and sorting move only slightly, to 0.114 and 0.051.

A common degree-5 polynomial raises worker variance relative to the additive model, from 0.478 to 0.495, while slightly lowering firm variance to 0.113 and sorting to 0.048. Interacting the polynomial basis with worker groups yields worker variance of 0.493, firm

identified in the presence of worker and year fixed effects. All variance components in Tables 3 and B5 are reported as shares of total log-wage variance on the same leave-match-out estimation set.

Table 3: Wage variance decomposition

Model	σ^2_α	σ^2_ψ	$\text{cov}(\alpha, \psi)$
Panel A: Parsimonious controls (Panel B sample)			
AKM baseline	0.5531	0.1359	0.0885
Linear additive controls	0.5338	0.1291	0.0816
Heterogeneous linear controls	0.5399	0.1290	0.0827
Nonlinear homogeneous controls	0.5509	0.1294	0.0841
Full model	0.5627	0.1288	0.0852
Panel B: Worker and firm-input controls			
AKM baseline	0.5531	0.1359	0.0885
Linear additive controls	0.4777	0.1153	0.0530
Heterogeneous linear controls	0.4546	0.1138	0.0510
Nonlinear homogeneous controls	0.4947	0.1134	0.0478
Full model	0.4926	0.1095	0.0495

Note: Entries report residualized leave-match-out KSS variance components as shares of total log-wage variance on the reported estimation sample: worker-effect variance (σ^2_α / var(y)), firm-effect variance (σ^2_ψ / var(y)), and their covariance (cov(α, ψ) / var(y)). Panel A is rerun on the same Panel B non-missing sample, so the two panels differ in controls rather than in sample composition. Panel B uses log employment and $\log(1 + x)$ for the two monetary inputs. Age is normalized at 50; the other continuous inputs are sample-centered, and all continuous inputs are scaled before constructing the polynomial basis. The table reports point estimates only. The specification sequence is: "Linear additive controls" uses the standard cubic age profile centered at 50 and linear firm inputs; "Heterogeneous linear controls" allows group-specific age deviations and firm-input slopes; "Nonlinear homogeneous controls" uses a common degree-5 polynomial basis; "Full model" interacts that basis with worker groups. Bias-corrected components need not sum to one.

variance of 0.110, and sorting of 0.049. The residualized-wage variance ratios are 0.827, 0.798, 0.832, and 0.827 for the additive, heterogeneous-linear, common-polynomial, and full specifications, respectively; the heterogeneous-linear specification has the smallest ratio.

What changes across specifications?. Employment, fixed assets, and intermediate inputs are economically meaningful firm-side controls, but most of their transformed variation is cross-sectional. The largest movement in Panel B comes from adding observed controls. Relative to the baseline, the linear additive specification lowers worker variance by 0.075, firm variance by 0.021, and sorting by 0.036. Relative to that additive specification, the common polynomial raises worker variance by 0.017 while lowering firm variance by 0.002 and sorting by 0.005. Adding group interactions to the polynomial changes the three components by only -0.002, -0.004, and $+0.002$. The ladder therefore distinguishes a substantial observables adjustment from the smaller reallocation generated by functional-form flexibility.

The Panel B estimates are stable across polynomial and Hermite bases. Their common specifications yield worker variance of 0.495 and 0.493, respectively, firm variance of 0.113 under both, and sorting of 0.048 under both. Their group-specific specifications also give nearly identical firm and sorting components. The cubic B-spline gives a common decomposition of 0.496, 0.110, and 0.047, and a group-specific decomposition of 0.476,

Table 4: Variance remaining after partialling out observables

Model	$\mathrm{var}(\tilde{y})$ / $\mathrm{var}(y)$
Panel A: Parsimonious controls (Panel B sample)	
AKM baseline	0.9985
Linear additive controls	0.9561
Heterogeneous linear controls	0.9642
Nonlinear homogeneous controls	0.9783
Full model	0.9913
Panel B: Worker and firm-input controls	
AKM baseline	0.9985
Linear additive controls	0.8271
Heterogeneous linear controls	0.7982
Nonlinear homogeneous controls	0.8319
Full model	0.8271

Note: Entries report $\mathrm{var}(\tilde{y})$ / $\mathrm{var}(y)$ on the reported leave-match-out estimation sample, where $\tilde{y}$ is the AKM input outcome after partialling out observables and year effects, before decomposing the remaining variation into worker, firm, and sorting components. This is a descriptive accounting statistic, not a bounded variance-explained share or the full residual from the joint worker-firm-controls-year regression. Values above one indicate that the observables-and-year partialling-out step amplifies variance under severe collinearity.

0.107, and 0.048. Across the three group-specific bases, firm variance therefore lies between 0.107 and 0.110, and sorting between 0.048 and 0.050; the worker component ranges from 0.476 to 0.493.

In Panel A, firm variance and sorting are also stable across bases, but the group-specific worker component ranges from 0.550 under Hermite to 0.563 under monomials. The monomial and Hermite regressions fit the same wage in this panel, so this difference is the constant-placement effect noted above: the two families split the same fitted variation differently between the age profile and the worker effects, rather than fitting the wage differently.

8.4. Interpreting the role of flexible controls

The within-panel ladder shows that adding observed characteristics materially changes the Panel B decomposition, while alternative functional forms produce smaller changes in its firm and sorting components. The bulk of the movement comes from including the firm inputs at all, not from how flexibly they enter: once they are in the model linearly, a degree-5 basis reallocates little. For these covariates in Portuguese data, the standard linear-in-logs adjustment is therefore close to adequate; the added flexibility mainly refines the worker component, which is the one component sensitive to the basis used for the group-specific controls. The full specification allows the return to age and the transformed firm inputs to differ across demographic groups (gender in Panel A; gender $\times$ education $\times$ qualification in Panel B).

Several caveats apply. First, this conclusion speaks only to heterogeneity along observed z_{it}. Match-specific premia or complementarities outside z_{it} can still matter (e.g., a manager–worker fit that raises pay on a particular team). Second, because the interaction groups are defined by worker categories rather than firm identifiers, the model does not estimate firm-specific pay schedules *per se*. Firms differ in their effective covariate adjustment only to the extent that they employ different mixes of worker types and have different levels of the continuous firm controls. Allowing firm-level interaction groups would require a separate design. Third, our decomposition is static: time-varying firm policies or match dynamics can exist without affecting the estimated variance components.

9. Conclusions

Standard AKM wage decompositions assume that observed covariates enter log wages linearly. This paper establishes when a leave-out estimator remains valid after replacing that restriction with a growing series approximation.

On the theory side, we developed a semiparametric leave-out estimator that replaces the linear control function with an unknown smooth function approximated by a growing basis. The estimator preserves the original AKM variance components while estimating a nonlinear control function. We established consistency and asymptotic normality under heteroskedasticity and many fixed effects, identifying the smoothness condition that separates quadratic estimands from the linear functionals studied in the existing semiparametric series literature. The required smoothness is stronger—the unknown function must have more continuous derivatives than continuous covariates—but is mild in typical wage applications.

On the empirical side, adding education, qualification, and transformed firm inputs lowers worker variance, firm variance, and sorting in the residualized KSS decomposition. Moving from linear to nonlinear controls produces smaller changes in firm variance and sorting; across the group-specific nonlinear bases, firm variance remains between 0.107 and 0.110 and sorting between 0.048 and 0.050. The worker component is more sensitive to the basis used for group-specific controls. In the parsimonious specification with only gender and age, the decomposition is stable across the specification ladder. These results make the ladder—and the distinction between adding observables and changing their functional form—central to empirical interpretation. They do not constitute a direct empirical implementation of the augmented-regressor correction developed in the theory.

Three limitations point to next steps. Our theory covers fixed-dimensional series designs. Extending the estimator to settings with growing continuous dimension, high-dimensional sparse structure, or machine-learning nuisance estimators would require new identifying structure and new computational arguments beyond the current leave-out linear-series setup. The interaction groups in our empirical specification are defined by worker categories; allowing firm-level interactions would permit firm-specific pay schedules, though at the cost of near-collinearity with firm fixed effects. Finally, the simulations show that finite-sample inference can under-cover even when point estimation is reliable, most clearly in the generic high-leverage design. The Portuguese-graph calibration reaches the opposite diagnostic extreme: its KSS point correction is nearly unbiased, but the feasible standard error is much too large after about half of the leave-out variance estimates are excluded. Improving confidence-interval coverage in these designs remains open.

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A. Proofs

This appendix collects the proofs. We first record the leave-one-out residual-variance identity used throughout, then prove the consistency and limiting-distribution results in the order they appear in the paper: Lemma 1, Lemma 2, Theorem 3, and Theorem 4.

Leave-one-out residual variance. The representation in (5) holds by applying the Sherman–Morrison formula to the leave-one-out estimator of γ. Denoting $S_{ww} := \sum_{i=1}^{n} w_i w_i'$, it holds that

$$\left(S_{ww} - w_i w_i'\right)^{-1} = S_{ww}^{-1} + \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1}}{1 - w_i' S_{ww}^{-1} w_i}.$$

Plugging the above into the definition of $\hat{\gamma}_{-i}$, we obtain

$$\begin{aligned}
\hat{\gamma}_{-i} &= \left(S_{ww}^{-1} + \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1}}{1 - w_i' S_{ww}^{-1} w_i}\right)\left(\sum_{i=1}^{n} w_i y_i - w_i y_i\right) \\
&= S_{ww}^{-1} \sum_{i=1}^{n} w_i y_i - S_{ww} w_i y_i + \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1} \sum_{i=1}^{n} w_i y_i}{1 - w_i' S_{ww}^{-1} w_i} - \frac{S_{ww}^{-1} w_i w_i' S_{ww}^{-1} w_i y_i}{1 - w_i' S_{ww}^{-1} w_i} \\
&= \hat{\gamma} + S_{ww}^{-1} w_i \left(\frac{w_i' \hat{\gamma}}{1 - w_i' S_{ww}^{-1} w_i} - y_i - \frac{w_i' S_{ww}^{-1} w_i y_i}{1 - w_i' S_{ww}^{-1} w_i}\right) \\
&= \hat{\gamma} + S_{ww}^{-1} w_i \left(\frac{w_i' \hat{\gamma} - y_i}{1 - w_i' S_{ww}^{-1} w_i}\right) \\
&= \hat{\gamma} - S_{ww}^{-1} w_i \frac{(y_i - w_i' \hat{\gamma})}{1 - w_i' S_{ww}^{-1} w_i}.
\end{aligned}$$

Substituting for $\hat{\gamma}_{-i}$ in (4), we have

$$\begin{aligned}
\hat{\sigma}_i^2 &= y_i \left(y_i - w_i' \hat{\gamma}_{-i}\right) \\
&= y_i \left(y_i - w_i' \left(\hat{\gamma} - S_{ww}^{-1} w_i \frac{(y_i - w_i' \hat{\gamma})}{1 - w_i' S_{ww}^{-1} w_i}\right)\right) \\
&= y_i \left(y_i - w_i' \hat{\gamma} + w_i' S_{ww}^{-1} w_i \frac{(y_i - w_i' \hat{\gamma})}{1 - w_i' S_{ww}^{-1} w_i}\right) \\
&= y_i \left((y_i - w_i' \hat{\gamma}) \left(\frac{1}{1 - w_i' S_{ww}^{-1} w_i}\right)\right) \\
&= \frac{y_i \hat{e}_i}{M_{W,ii}}.
\end{aligned}$$

Proof of Lemma 1. The variance of $\hat{\beta}$ is

$$
\begin{aligned}
\mathrm{var}[\hat{\beta}] &= \mathrm{var}\left[S_{xx}^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n}M_{ij}x_i f(z_j) + S_{xx}^{-1}\sum_{i=1}^{n}\sum_{j=1}^{n}M_{ij}x_i e_j\right] \\
&= S_{xx}^{-1}\,\mathrm{var}\left[\sum_{i=1}^{n}\sum_{j=1}^{n}M_{ij}x_i f(z_j) + \sum_{i=1}^{n}\sum_{j=1}^{n}M_{ij}x_i e_j\right]S_{xx}^{-1} \\
&= S_{xx}^{-1}\left(\sum_{i=1}^{n}\sum_{j=1}^{n}M_{ij}^2 x_i x_i' \sigma_j^2\right)S_{xx}^{-1}
\end{aligned}
$$

under the assumption of fixed x_i and z_i.

First, rewrite the difference between the estimator and the estimand as

$$
\begin{aligned}
\hat{\theta} - \theta &= \hat{\beta}' A \hat{\beta} - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} y_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a y_b - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2.
\end{aligned}
$$

Given a data-generating process for y_i, we can expand $\hat{\theta} - \theta$ further as

$$
\begin{aligned}
\hat{\theta} - \theta &= \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}\left(x_j'\beta + f(z_j) + e_j\right) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a \left(x_b'\beta + f(z_b) + e_b\right) - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 \\
&= \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_j'\beta x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a x_b'\beta + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} f(z_j) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a f(z_b) \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} e_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a e_b + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_j'\beta x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a f(z_b) \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} f(z_j) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a x_b'\beta + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} x_j'\beta x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a e_b \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} e_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a x_b'\beta + \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} f(z_j) x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a e_b \\
&+ \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij} e_j x_i' S_{xx}^{-1} A S_{xx}^{-1} \sum_{a=1}^{n}\sum_{b=1}^{n} M_{ab} x_a f(z_b) - \beta' A \beta - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2.
\end{aligned}
$$

Next, we use the definition $B_{i\ell} := \sum_{j=1}^{n} M_{ij} x_j' S_{xx}^{-1} A S_{xx}^{-1} \sum_{j=1}^{n} M_{\ell j} x_j$, so that

$$
\begin{aligned}
\hat{\theta} - \theta &= \sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 + \sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} e_i^2 - \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 \\
&\quad + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell'\beta f(z_i) + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell'\beta e_i + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} e_\ell f(z_i).
\end{aligned}
$$

Finally, rearranging, we have

$$\begin{aligned}\hat{\theta} - \theta = &\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 + \sum_{i=1}^{n}\sum_{\ell\neq i} B_{i\ell} e_i e_\ell + \sum_{i=1}^{n} B_{ii}(e_i^2 - \hat{\sigma}_i^2) \\ &+ 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta f(z_i) + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta e_i + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} e_\ell f(z_i).\end{aligned} \tag{A1}$$

In general, it holds that

$$\mathbb{E}[|\hat{\theta} - \theta|^2] = |\mathbb{E}[\hat{\theta} - \theta]|^2 + \text{trace}(\text{var}[\hat{\theta} - \theta]).$$

To show that $\hat{\theta}$ is consistent for θ, we need to show that the bias and the variance of the difference in (A1) goes to zero. The main idea is to compute bounds on each term and their variances, and show that these bounds are asymptotically negligible. Then, convergence in the quadratic mean would imply convergence in probability.

Applying expectations on both sides, and using independence and mean-zero properties of errors, we have

$$\mathbb{E}[\hat{\theta} - \theta] = \sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 + 2\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} x_\ell' \beta f(z_i) + \sum_{i=1}^{n} B_{ii}\mathbb{E}[\sigma_i^2 - \hat{\sigma}_i^2].$$

We now prove that each term goes to zero in probability as $n \to \infty$, $k \to \infty$, and $p \to \infty$.

Denote $B := (B_{i\ell})_{i,\ell=1}^{n} \in \mathbb{R}^{n\times n}$, and $F := (f(z_1), \ldots, f(z_n))' \in \mathbb{R}^n$. Then

$$\begin{aligned}\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 &= F'BF \\ &= F'MXS_{xx}^{-1}AS_{xx}^{-1}X'MF \\ &= F'MXS_{xx}^{-1}A^{1/2}A^{1/2}S_{xx}^{-1}X'MF \\ &= (F'MXS_{xx}^{-1}A^{1/2})^2,\end{aligned}$$

where we use $A = A^{1/2}A^{1/2}$ because A is symmetric. By the Markov inequality, assumptions on $\mathcal{F}$, M being idempotent, and the Cauchy-Schwarz inequality,

$$\begin{aligned}\left\| \frac{1}{n} F'MXS_{xx}^{-1}A^{1/2} \right\| &\leq \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \cdot \text{trace}\left(\frac{1}{n}A^{1/2}S_{xx}^{-1}X'MXS_{xx}^{-1}A^{1/2}\right)^{1/2} \\ &= \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \cdot \text{trace}\left(\frac{1}{n}A^{1/2}S_{xx}^{-1}A^{1/2}\right)^{1/2} \\ &= \mathcal{O}(k^{-\alpha_f}\sqrt{p/n}).\end{aligned}$$

Thus, using the Assumption 1 and 3, we have

$$\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell} f(z_i)^2 = \mathcal{O}(k^{-2\alpha_f} np) = \mathcal{O}(k^{-\alpha_f} n) \to 0.$$

Now, for the second term in the main decomposition,

$$\begin{aligned}\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell}x_\ell'\beta f(z_i) &= F'BX\beta \\ &= F'MXS_{xx}^{-1}AS_{xx}^{-1}X'MX\beta \\ &= F'MXS_{xx}^{-1}A\beta.\end{aligned}$$

Given that $\beta' AS_{xx}^{-1}A\beta = \mathcal{O}(1)$, by the Markov inequality, assumption on $\mathcal{F}$, M being idempotent, and the Cauchy-Schwarz inequality we have

$$\begin{aligned}\left\|\frac{1}{n}F'MXS_{xx}^{-1}A\beta\right\| &\leq \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2}\cdot\text{trace}\left(\frac{1}{n}\beta' AS_{xx}^{-1}X'MXS_{xx}^{-1}A\beta\right)^{1/2} \\ &= \text{trace}\left(\frac{1}{n}F'MF\right)^{1/2}\cdot\text{trace}\left(\frac{1}{n}\beta' AS_{xx}^{-1}A\beta\right)^{1/2} \\ &= \mathcal{O}(k^{-\alpha_f}/\sqrt{n}).\end{aligned}$$

So that using the Assumption 1 and 3, we have

$$\sum_{i=1}^{n}\sum_{\ell=1}^{n} B_{i\ell}x_\ell'\beta f(z_i) = \mathcal{O}(k^{-\alpha_f}\sqrt{n}) \to 0.$$

To bound the third term, we should show that

$$\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2] = \left(x_i'\beta + f(z_i)\right)\left(x_i'\mathbb{E}[\beta - \hat{\beta}_{-i}] + \mathbb{E}[f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}]\right)$$

goes in probability to zero. This is equivalent to bounding the bias of the estimator of the slope coefficient and the bias of the function approximation.

The bias of the slope estimator is

$$\mathbb{E}[\hat{\beta} - \beta] = \mathbb{E}\left[\left(\frac{1}{n}X'MX\right)^{-1}\frac{1}{n}X'MF\right].$$

By the Markov inequality, assumption on $\mathcal{F}$, M being idempotent, and the Cauchy-Schwarz inequality we have that:

$$\begin{aligned}\left\|\frac{1}{n}X'MF\right\| &\leq \text{trace}\left(\frac{1}{n}X'MX\right)^{1/2}\cdot\text{trace}\left(\frac{1}{n}F'MF\right)^{1/2} \\ &= \mathcal{O}(k^{-\alpha_f}\sqrt{p/n}),\end{aligned}$$

and, similarly,

$$\left\|\frac{1}{n}X'MX\right\| = \mathcal{O}(p/n),$$

so that using Assumption 1 and 3, we ultimately have:

$$\mathbb{E}[\|\hat{\beta} - \beta\|] = \mathcal{O}(\sqrt{n}/k^{\alpha_f}\sqrt{p}) \to 0.$$

Because of the assumption on the functional class $\mathcal{F}$, we can bound the bias of the function approximation using Jensen's inequality for convex $x \mapsto x^2$,

$$\begin{aligned}\mathbb{E}[|f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}|] &\leq \mathbb{E}[|f(z_i) - p_k(z_i)'\hat{\alpha}_{-i}|^2]^{1/2} \\ &\leq (Ck^{-2\alpha_f})^{1/2} = \mathcal{O}_p(k^{-\alpha_f}),\end{aligned}$$

so that:

$$\mathbb{E}[|\hat{\sigma}_i^2 - \sigma_i^2|] = \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p}) \xrightarrow{p} 0.$$

From this it follows that:

$$\mathbb{E}[|\hat{\theta} - \theta|] \xrightarrow{p} 0.$$

We next turn our attention to bounding the variance. Let matrix

$$\tilde{A} := S_{xx}^{-1/2} A S_{xx}^{-1/2},$$

and let $\lambda_1, \ldots, \lambda_r$ be its nonzero eigenvalues. We assume that $\lambda_1^2 \geq \ldots \geq \lambda_r^2$, and that each eigenvalue appears as many times as its algebraic multiplicity. Under these assumptions, we can spectrally decompose $\tilde{A} = QDQ'$, where Q is a matrix of orthonormal vectors and $D = \text{diag}(\lambda_1, \ldots, \lambda_r)$.

The variance of $2\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell} x_\ell' \beta e_i$ is

$$\begin{aligned}4\sum_{i=1}^n \left(\sum_{\ell=1}^n B_{i\ell} x_\ell' \beta\right)^2 \sigma_i^2 &\leq \max_i \sigma_i^2 \beta' X' B^2 X \beta = \max_i \sigma_i^2 \beta' A S_{xx}^{-1} A \beta \\ &\leq \max_i \sigma_i^2 \lambda_1 \theta = o(1).\end{aligned}$$

To explain why the last inequality holds, define $\tilde{\beta} = S_{xx}^{1/2}\beta$ so that $\beta = S_{xx}^{-1/2}\tilde{\beta}$,

$$\beta' A S_{xx}^{-1} A \beta = \tilde{\beta}' S_{xx}^{-1/2} A S_{xx}^{-1/2} \tilde{\beta} = \tilde{\beta}' (S_{xx}^{-1/2} A S_{xx}^{-1/2})^2 \tilde{\beta} = \tilde{\beta}' \tilde{A}^2 \tilde{\beta},$$

and

$$\theta = \tilde{\beta}' S_{xx}^{-1/2} A S_{xx}^{-1/2} \tilde{\beta} = \tilde{\beta}' \tilde{A} \tilde{\beta}.$$

Then, using the Rayleigh quotient argument, it holds that

$$\frac{\tilde{\beta}' \tilde{A}^2 \tilde{\beta}}{\tilde{\beta}' \tilde{A} \tilde{\beta}} \leq \lambda_{\max}(\tilde{A}),$$

because $\tilde{A}$ is positive semi-definite, $\tilde{\beta} \neq 0$, and $\lambda_{\max}(\tilde{A})$ is the largest eigenvalue of $\tilde{A}$. From this it follows that we can bound

$$\beta' A S_{xx}^{-1} A \beta \leq \lambda_1 \theta,$$

use assumptions $\theta = \mathcal{O}(1)$, $\lambda_1 \leq \text{trace}(\tilde{A}^2)^{1/2} = o(1)$, and the fact that the variance is bounded. The conclusion then follows.

The variance of $2\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell} e_\ell f(z_i)$ is

$$4\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 \sigma_\ell^2 f(z_i)^2 \leq \max_i 4\sigma_i^2 f(z_i)^2 \sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 = \max_i 4\sigma_i^2 f(z_i)^2 \text{trace}(\tilde{A}^2) = o(1),$$

which follows since

$$\begin{aligned}
\sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 = \|B\|_F^2 = \text{trace}(B'B) & \\
&= \text{trace}(MXS_{xx}^{-1}AS_{xx}^{-1}AS_{xx}^{-1}X'M) \\
&= \text{trace}(MXS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}X'M) \\
&= \text{trace}(S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}X'MXS_{xx}^{-1/2}) \\
&= \text{trace}(S_{xx}^{-1/2}AS_{xx}^{-1/2}S_{xx}^{-1/2}AS_{xx}^{-1/2}) = \text{trace}(\tilde{A}^2),
\end{aligned}$$

and we use the assumptions as above but now, instead of the bounded variance, we assume that $\mathcal{F}$ is a class of bounded functions so that $\max|f(z_i)| < C$ for $i = 1, \ldots, n$ for some absolute constant C (this is implied by the main assumption on $\mathcal{F}$).

Because $B_{i\ell}^2 = B_{\ell i}^2$ for any i, ℓ, the variance of $\sum_{i=1}^n \sum_{\ell \neq i} B_{i\ell} e_i e_\ell$ is

$$2\sum_{i=1}^n \sum_{\ell \neq i} B_{i\ell}^2 \sigma_i^2 \sigma_\ell^2 \leq \max_i 2\sigma_i^4 \sum_{i=1}^n \sum_{\ell=1}^n B_{i\ell}^2 = \max_i 2\sigma_i^4 \text{trace}(\tilde{A}^2) = o(1).$$

To compute the variance of $\sum_{i=1}^n B_{ii}(e_i^2 - \hat{\sigma}_i^2)$, we represent the leave-one-out variance estimator as

$$\begin{aligned}
\hat{\sigma}_i^2 &= y_i(y_i - w_i'\hat{\gamma}_{-i}) = y_i M_{W,ii}^{-1}(y_i - w_i'\hat{\gamma}) \\
&= y_i M_{W,ii}^{-1}\hat{e}_i = y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} y_\ell \\
&= y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} \left(x_\ell'\beta + f(z_\ell) + e_\ell\right) \\
&= y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} \left(x_\ell'\beta + f(z_\ell)\right) + y_i M_{W,ii}^{-1} \sum_{\ell=1}^n M_{W,i\ell} e_\ell.
\end{aligned}$$

Thus, the variance is now expressed as:

$$\begin{aligned}
\sum_{i=1}^n \left(\sum_{\ell=1}^n M_{W,\ell\ell}^{-1} B_{\ell\ell} M_{W,i\ell} \left(x_\ell'\beta + f(z_\ell)\right) \right)^2 \sigma_i^2 + 2\sum_{i=1}^n \sum_{\ell \neq i} M_{W,ii}^{-2} B_{ii}^2 M_{W,i\ell}^2 \sigma_i^2 \sigma_\ell^2 & \\
\leq \frac{1}{c^2} \max_i \sigma_i^2 \max_i \left(x_i'\beta + f(z_i)\right)^2 \sum_{i=1}^n B_{ii}^2 + \frac{2}{c} \max_i \sigma_i^4 \sum_{i=1}^n B_{ii}^2 = o(1)&,
\end{aligned}$$

because $\min_i M_{W,ii} \geq c > 0$, $\sum_{i=1}^n B_{ii}^2 \leq \text{trace}(\tilde{A}^2) = o(1)$, and $\max_i \left(x_i'\beta + f(z_i)\right)^2 \leq 2\max_i(x_i'\beta)^2 + 2\max_i f(z_i)^2 = \mathcal{O}(1)$.

Because we have that

$$\mathbb{E}[\hat{\theta} - \theta] \xrightarrow{p} 0, \quad \text{var}[\hat{\theta} - \theta] \xrightarrow{p} 0,$$

the proposed estimator $\hat{\theta}$ is consistent. $\square$

Proof of Lemma 2. We prove the result by considering a second-order approximation of $\hat{\theta}_{\text{JLA}} - \hat{\theta}$ around $\hat{a}_i := (1 - P_{W,ii})^{-1}(\hat{P}_{W,ii} - P_{W,ii})$ as

$$(\hat{\theta}_{\text{JLA}} - \hat{\theta})_2 := \sum_{i=1}^{n} \hat{\sigma}_i^2 \left(B_{ii} - \hat{B}_{ii} - \hat{B}_{ii}\hat{a}_i - \hat{B}_{ii} \left(\hat{a}_i^2 - \frac{1}{m} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} \right) \right),$$

and an approximation error that we show to be negligible,

$$\text{AE}_2 := \sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} \left(\frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2 - (3P_{W,ii}^3 + P_{W,ii}^2)(1 - \hat{a}_i)^2}{(1 - \hat{a}_i)^2(1 - P_{W,ii})} - \frac{\hat{a}_i^3}{1 - \hat{a}_i} \right).$$

To decompose $\hat{\theta}_{\text{JLA}} - \hat{\theta} = (\hat{\theta}_{\text{JLA}} - \hat{\theta})_2 + \text{AE}_2$, note that

$$\begin{aligned}
\hat{\sigma}_{i,\text{JLA}}^2 &= \frac{y_i(y_i - w_i'\hat{\gamma})}{1 - \hat{P}_{W,ii}} \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right) \\
&= \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \hat{\sigma}_i^2 \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right),
\end{aligned}$$

so that

$$\begin{aligned}
\hat{\theta}_{\text{JLA}} - \hat{\theta} &= \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 - \hat{B}_{ii}\hat{\sigma}_{i,\text{JLA}}^2 \\
&= \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 - \hat{B}_{ii}\hat{\sigma}_i^2 \left(1 - \frac{1}{m} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} \right) \\
&= \sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 + \hat{B}_{ii}\hat{\sigma}_i^2 \left(\frac{1}{m} \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} - \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \right).
\end{aligned}$$

Add and subtract $m^{-1}\hat{\sigma}_i^2 \hat{B}_{ii}(1 - P_{W,ii})^{-1}(3P_{W,ii}^3 + P_{W,ii}^2)$ to obtain

$$\begin{aligned}
\hat{\theta}_{\text{JLA}} - \hat{\theta} = \sum_{i=1}^{n} & B_{ii}\hat{\sigma}_i^2 + \frac{1}{m}\hat{\sigma}_i^2 \hat{B}_{ii} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} + \hat{B}_{ii}\hat{\sigma}_i^2 \left(\frac{1}{m} \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2}{1 - \hat{P}_{W,ii}} - \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \right) \\
& - \frac{1}{m}\hat{\sigma}_i^2 \hat{B}_{ii} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} \\
= & \sum_{i=1}^{n} \hat{\sigma}_i^2 \left(B_{ii} + \frac{1}{m} \hat{B}_{ii} \frac{3P_{W,ii}^3 + P_{W,ii}^2}{1 - P_{W,ii}} \right) \\
& + \sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} \left(\frac{1}{m} \frac{(1 - P_{W,ii})^2(3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2) - (1 - \hat{P}_{W,ii})^2(3P_{W,ii}^3 + P_{W,ii}^2)}{(1 - \hat{P}_{W,ii})^2(1 - P_{W,ii})} - \frac{1 - P_{W,ii}}{1 - \hat{P}_{W,ii}} \right),
\end{aligned}$$

and using $(1-\hat{P}_{W,ii})^{-1}(1-P_{W,ii}) = (1-\hat{a}_i)^{-1}$, and expanding up to the third order as

$$\frac{1-P_{W,ii}}{1-\hat{P}_{W,ii}} = 1 + \frac{1-P_{W,ii}}{1-\hat{P}_{W,ii}}\hat{a}_i$$
$$= 1 + \hat{a}_i + \hat{a}_i^2 + \frac{1-P_{W,ii}}{1-\hat{P}_{W,ii}}\hat{a}_i^3,$$

we have that

$$\begin{aligned}\hat{\theta}_{\text{JLA}} - \hat{\theta} &= \sum_{i=1}^n \hat{\sigma}_i^2 \left(B_{ii} + \frac{1}{m}\hat{B}_{ii}\frac{3P_{W,ii}^3 + P_{W,ii}^2}{1-P_{W,ii}} \right) \\ &\quad + \sum_{i=1}^n \hat{\sigma}_i^2 \hat{B}_{ii} \left(\frac{1}{m}\frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2 - (1-\hat{a}_i)^2(3P_{W,ii}^3 + P_{W,ii}^2)}{(1-\hat{a}_i)^2(1-P_{W,ii})} - \left(1 + \hat{a}_i + \hat{a}_i^2 + \frac{\hat{a}_i^3}{1-\hat{a}_i}\right)\right) \\ &= (\hat{\theta}_{\text{JLA}} - \hat{\theta})_2 + \text{AE}_2.\end{aligned}$$

To describe the bias, we note that $\hat{P}_{W,ii}$, $\hat{B}_{ii}$, and $\hat{\sigma}_i^2$ are independent of each other, $\mathbb{E}[\hat{P}_{W,ii}] = P_{W,ii}$, $\mathbb{E}[\hat{B}_{ii}] = B_{ii}$, $\mathbb{E}[\hat{\sigma}_i^2] = \sigma_i^2 + \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p})$, and using properties of the Rademacher random variables,

$$\text{var}[\hat{a}_i] = \frac{2}{m}\frac{P_{W,ii} - \sum_{\ell=1}^n P_{W,i\ell}^4}{(1-P_{W,ii})^2} = \frac{1}{m}\frac{3P_{W,ii}^3 + P_{W,ii}^2}{1-P_{W,ii}} + \frac{P_{W,ii}(1-P_{W,ii})^2 - 2\sum_{\ell\neq i}^n P_{W,i\ell}^4}{m(1-P_{W,ii})^2}.$$

Therefore, in total we have:

$$\mathbb{E}[(\hat{\theta}_{\text{JLA}} - \hat{\theta})_2] = -\sum_{i=1}^n \sigma_i^2 B_{ii}\left(\text{var}[\hat{a}_i] - \frac{1}{m}\frac{3P_{W,ii}^3 + P_{W,ii}^2}{1-P_{W,ii}}\right) + \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p}),$$

or, assuming $k \to \infty$,

$$\mathbb{E}[(\hat{\theta}_{\text{JLA}} - \hat{\theta})_2] = \text{B}_m + o(1), \quad \text{B}_m := \sum_{i=1}^n B_{ii}\sigma_i^2\left(\frac{2\sum_{\ell\neq i} P_{W,i\ell}^4 - P_{W,ii}^2(1-P_{W,ii})^2}{m(1-P_{W,ii})^2}\right).$$

Focusing on the variance next, denote $y := (y_1, \ldots, y_n)'$, so that

$$\begin{aligned}\text{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2(B_{ii} - \hat{B}_{ii})\right] &= \mathbb{E}\left[\text{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\right]\middle| y\right] + \text{var}\left[\mathbb{E}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\right]\middle| y\right] = \mathbb{E}\left[\text{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\right]\middle| y\right] \\ &\leq 2m^{-1}\sum_{i=1}^n\sum_{\ell=1}^n B_{i\ell}^2\mathbb{E}[\hat{\sigma}_i^2\hat{\sigma}_\ell^2] = \mathcal{O}\left(m^{-1}\text{trace}(\tilde{A}^2)\right),\end{aligned}$$

$$\begin{aligned}\text{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\hat{a}_i\right] &= \mathbb{E}\left[\text{var}\left[\sum_{i=1}^n \hat{\sigma}_i^2\hat{B}_{ii}\hat{a}_i\middle| y, R_B\right]\right] \leq 2m^{-1}\sum_{i=1}^n\sum_{\ell=1}^n P_{W,i\ell}^2\frac{\mathbb{E}[\hat{B}_{ii}\hat{B}_{\ell\ell}]\mathbb{E}[\hat{\sigma}_i^2\hat{\sigma}_\ell^2]}{(1-P_{W,ii})(1-P_{W,\ell\ell})} \\ &= \mathcal{O}\left(m^{-1}\text{trace}(\tilde{A}^2) + m^{-2}\text{trace}(\tilde{A}_1^2)^{1/2}\text{trace}(\tilde{A}_2^2)^{1/2}\right)\end{aligned}$$

for $\tilde{A}_k := S_{xx}^{-1/2} A_k' A_k S_{xx}^{-1/2}$ for $k = 1, 2$. Regarding the ensuing terms, it holds that:

$$\begin{aligned}\mathrm{var}\left[\sum_{i=1}^{n} \hat{\sigma}_i^2 \hat{B}_{ii} \left(\hat{a}_i^2 - \mathrm{var}[\hat{a}_i]\right)\right] &= \sum_{i=1}^{n}\sum_{\ell=1}^{n} \mathbb{E}[\hat{B}_{ii}\hat{B}_{\ell\ell}]\mathbb{E}[\hat{\sigma}_i^2\hat{\sigma}_\ell^2]\,\mathrm{cov}[\hat{a}_i^2, \hat{a}_\ell^2] \\ &= \mathcal{O}\left(m^{-2}\mathrm{trace}(\tilde{A}^2) + m^{-3}\mathrm{trace}(\tilde{A}_1^2)^{1/2}\mathrm{trace}(\tilde{A}_2^2)^{1/2}\right),\end{aligned}$$

$$\mathrm{var}\left[\sum_{i=1}^{n} \hat{\sigma}_i^2 (\hat{B}_{ii} - B_{ii}) \frac{2\sum_{\ell\neq i}^{n} P_{W,i\ell}^4 - P_{W,ii}(1-P_{W,ii})^2}{m(1-P_{W,ii})^2}\right] = \mathcal{O}\left(m^{-3}\mathrm{trace}(\tilde{A}^2)\right),$$

$$\mathrm{var}\left[\sum_{i=1}^{n} B_{ii}\left(\hat{\sigma}_i^2 - \sigma_i^2\right) \frac{2\sum_{\ell\neq i}^{n} P_{W,i\ell}^4 - P_{W,ii}(1-P_{W,ii})^2}{m(1-P_{W,ii})^2}\right] = \mathcal{O}\left(m^{-2}\,\mathrm{var}[\hat{\theta}]\right).$$

Because $\mathrm{trace}(\tilde{A}^2) = \mathcal{O}(\mathrm{var}[\hat{\theta}])$ and $m^{-4}\,\mathrm{var}[\hat{\theta}]^{-2}\,\mathrm{var}[\hat{\theta}_1]\,\mathrm{var}[\hat{\theta}_2] = o(1)$, it can be ultimately established that $\mathrm{var}[\hat{\theta}]^{-1/2}\left((\hat{\theta}_{\mathrm{JLA}} - \hat{\theta})_2 - \mathrm{B}_m\right) = o_p(1)$.

Using that $\mathbb{E}[\hat{a}_i^3] = \mathcal{O}(m^{-2})$, $\mathbb{E}[\hat{a}_i^4] = \mathcal{O}(m^{-2})$, and $\max_i |\hat{a}_i| = o_p(\log n/\sqrt{m})$, the terms in the approximation error are as follows:

$$\begin{aligned}\sum_{i=1}^{n}\hat{\sigma}_i^2\hat{B}_{ii}\hat{a}_i^3 + \sum_{i=1}^{n}\hat{\sigma}_i^2\hat{B}_{ii}\hat{a}_i^4 &= m^{-2}\mathcal{O}_p\left(\mathbb{E}[\hat{\theta}_{1,\mathrm{PI}} - \theta_1] + \mathbb{E}[\hat{\theta}_{2,\mathrm{PI}} - \theta_2]\right), \\ \sum_{i=1}^{n}\hat{\sigma}_i^2\hat{B}_{ii}\frac{\hat{a}_i^5}{1-\hat{a}_i} &= \frac{\log n}{m^{5/4}}\mathcal{O}_p\left(\mathbb{E}[\hat{\theta}_{1,\mathrm{PI}} - \theta_1] + \mathbb{E}[\hat{\theta}_{2,\mathrm{PI}} - \theta_2]\right), \\ \frac{1}{m}\sum_{i=1}^{n}\hat{\sigma}_i^2\hat{B}_{ii}&\frac{3\hat{P}_{W,ii}^3 + \hat{P}_{W,ii}^2 - (3P_{W,ii}^3 + P_{W,ii}^2)(1-\hat{a}_i)^2}{(1-\hat{a}_i)^2(1-P_{W,ii})} \\ &= \left(m^{-2} + \frac{\log n}{p^{5/4}}\right)\mathcal{O}_p\left(\mathbb{E}[\hat{\theta}_{1,\mathrm{PI}} - \theta_1] + \mathbb{E}[\hat{\theta}_{2,\mathrm{PI}} - \theta_2]\right).\end{aligned}$$

□

Proof of Theorem 3. Representation in the theorem holds because

$$\sum_{\ell=1}^{r}\lambda_\ell\hat{b}_\ell^2 = \hat{\beta}' S_{xx}^{1/2} Q D Q' S_{xx}^{1/2}\hat{\beta} = \hat{\beta}' S_{xx}^{1/2} S_{xx}^{-1/2} A S_{xx}^{-1/2} S_{xx}^{1/2}\hat{\beta} = \hat{\beta}' A\hat{\beta},$$

and

$$\sum_{i=1}^{n} B_{ii}\hat{\sigma}_i^2 = \mathrm{trace}(A\widehat{\mathrm{var}}[\hat{\beta}]) = \mathrm{trace}(D\widehat{\mathrm{var}}[\hat{b}]) = \sum_{\ell=1}^{r}\lambda_\ell\widehat{\mathrm{var}}[\hat{b}_\ell].$$

We prove the theorem in three steps.
Approximation. Equivalently, represent $\hat{\theta}$ as

$$\hat{\theta} = \sum_{\ell=1}^{r}\lambda_\ell\left(\hat{b}_\ell^2 - \mathrm{var}[\hat{b}_\ell]\right) + \sum_{i=1}^{n} B_{ii}(\sigma_i^2 - \hat{\sigma}_i^2),$$

and below we show that the second term is asymptotically dominated by the variance of the estimator, $\mathrm{var}[\hat{\theta}]$. Given that, the asymptotic distribution of $\hat{\theta}$ is then driven by the joint distribution of random vector $\hat{b}$.

We need to show that the second term is asymptotically mean-zero and is of smaller order than variance of $\hat{\theta}$. The first claim is immediate because we have shown in Lemma 1 that

$$\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2] = \mathcal{O}_p(\sqrt{n}/k^{\alpha_f}\sqrt{p}) \xrightarrow{p} 0,$$

so that

$$\mathbb{E}\left[\sum_{i=1}^n B_{ii}(\sigma_i^2 - \hat{\sigma}_i^2)\right] \xrightarrow{p} 0.$$

Now, rewrite

$$\begin{aligned}\sum_{i=1}^n B_{ii}(\hat{\sigma}_i^2 - \sigma_i^2) &= \sum_{i=1}^n B_{ii} M_{W,ii}^{-1} x_i'\beta \sum_{\ell=1}^n M_{W,i\ell} e_\ell + \sum_{i=1}^n (e_i^2 - \sigma_i^2) \\ &+ \sum_{i=1}^n B_{ii} M_{W,ii}^{-1} f(z_i) \sum_{\ell=1}^n M_{W,i\ell} e_\ell + \sum_{i=1}^n B_{ii} M_{W,ii}^{-1} \sum_{\ell \neq i} M_{W,i\ell} e_i e_\ell.\end{aligned}$$

The variances of the first and the third terms are

$$\sum_{\ell=1}^n \sigma_\ell^2 \left(\sum_{i=1}^n M_{W,i\ell} B_{ii} M_{W,ii}^{-1} x_i'\beta\right)^2 \leq \max_i \sigma_i^2 \sum_{i=1}^n B_{ii}^2 M_{W,ii}^{-2} (x_i'\beta)^2 \leq \max_i \sigma_i^2 \max_i (x_i'\beta)^2 M_{W,ii}^{-2} \sum_{i=1}^n B_{ii}^2,$$

$$\sum_{\ell=1}^n \sigma_\ell^2 \left(\sum_{i=1}^n M_{W,i\ell} B_{ii} M_{W,ii}^{-1} f(z_i)\right)^2 \leq \max_i \sigma_i^2 \sum_{i=1}^n B_{ii}^2 M_{W,ii}^{-2} f(z_i)^2 \leq \max_i \sigma_i^2 \max_i f(z_i)^2 M_{W,ii}^{-2} \sum_{i=1}^n B_{ii}^2,$$

and of the second and the fourth

$$\sum_{i=1}^n B_{ii}^2 \operatorname{var}[e_i^2] \leq \max_i \mathbb{E}[e_i^4] \sum_{i=1}^n B_{ii}^2,$$

$$\sum_{i=1}^n \sum_{\ell \neq i} (B_{ii}^2 M_{W,ii}^{-2} + B_{ii} M_{W,ii}^{-1} B_{\ell\ell} M_{W,\ell\ell}^{-1}) M_{W,i\ell}^2 \sigma_i^2 \sigma_\ell^2 \leq 2 \max_i \sigma_i^4 M_{W,ii}^{-2} \sum_{i=1}^n B_{ii}^2.$$

Because each variance is bounded by $C \sum_{i=1}^n B_{ii}^2$, to show that it is of smaller order than the variance of $\hat{\theta}$, we need $\operatorname{var}[\hat{\theta}]^{-1} \sum_{i=1}^n B_{ii}^2 = o(1)$. It holds because

$$\operatorname{var}[\hat{\theta}]^{-1} \sum_{i=1}^n B_{ii}^2 \leq \max_i v_i' v_i \operatorname{var}[\hat{\theta}]^{-1} \sum_{\ell=1}^r \lambda_\ell^2 \leq \max_i v_i' v_i \max_i \sigma_i^{-4} = o(1).$$

Variance estimator consistency. Now we show that the variance estimator is consistent, i.e. $\operatorname{var}[\hat{b}]^{-1} \widehat{\operatorname{var}}[\hat{b}] \xrightarrow{p} I_r$. For it we need to show that

$$\operatorname{var}[\vartheta'\hat{b}]^{-1} \left(\widehat{\operatorname{var}}[\vartheta'\hat{b}] - \operatorname{var}[\vartheta'\hat{b}]\right) = o_p(1), \quad \vartheta \in \mathbb{R}^r, \quad \vartheta'\vartheta = 1$$

for some non-random ϑ. Rewrite the expression above as

$$\delta(\vartheta) := \sum_{i=1}^n v_i(\vartheta)(\hat{\sigma}_i^2 - \sigma_i^2), \tag{A2}$$

where

$$v_i(\vartheta) := \frac{(\vartheta' v_i)^2}{\sum_{i=1}^n \sigma_i^2 (\vartheta' v_i)^2}.$$

We know that $\mathbb{E}[\delta(\vartheta)] = o_p(1)$ because by the triangle inequality, the Cauchy-Schwarz inequality, and $|\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2]| = o_p(1)$,

$$|v(\vartheta)\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2]| \leq (v(\vartheta)^2)^{1/2} \cdot \left(|\mathbb{E}[\hat{\sigma}_i^2 - \sigma_i^2]|^2\right)^{1/2} = o_p(1)$$

for $i = 1, \ldots, n$. The variance of $\delta(\vartheta)$ is

$$\begin{aligned}
\sum_{i=1}^n \delta(\vartheta)^2 \operatorname{var}[\hat{\sigma}_i^2] &\leq \sum_{i=1}^n v_i(\vartheta)^4 \\
&\leq \max_i \sigma_i^{-4} \max_i v_i v_i' \frac{\vartheta'\vartheta}{\sum_{i=1}^n v_i v_i' \vartheta'\vartheta} \\
&= \max_i \sigma_i^{-4} \max_i v_i v_i' = o(1)
\end{aligned}$$

because $\max_i v_i v_i' = o(1)$ by assumption.

Asymptotic normality. Our objective is to prove that

$$\operatorname{var}[\vartheta'\hat{b}]^{-1/2} \left(\vartheta'\hat{b} - \vartheta' b\right) \xrightarrow{d} \mathcal{N}(0,1),$$

where $\hat{b} := Q' S_{xx}^{1/2} \hat{\beta}$, and $b := Q' S_{xx}^{1/2} \beta$. Lyapunov's condition implies that it is sufficient to show that

$$\operatorname{var}[\vartheta'\hat{b}]^{-2} \sum_{i=1}^n \mathbb{E}\left[\left(\vartheta'(\hat{b} - b)\right)^4\right] = o_p(1).$$

Because we have that

$$\begin{aligned}
\vartheta'(\hat{b} - b) &= \vartheta'(Q' S_{xx}^{1/2} \hat{\beta} - Q' S_{xx}^{1/2} \beta) \\
&= \sum_{i=1}^n \vartheta' v_i \left(f(z_i) + e_i\right),
\end{aligned}$$

and $(a+b)^4 \leq C(a^4 + b^4)$ for some constant C, Lyapunov's condition is equivalent to

$$\operatorname{var}[\vartheta'\hat{b}]^{-2} \sum_{i=1}^n C\left(f(z_i)^4 + \mathbb{E}[e_i^4]\right) \cdot (\vartheta' v_i)^4 = o_p(1).$$

It holds because $\max_i |f(z_i)| = \mathcal{O}(1)$, and $\max_i \mathbb{E}[e_i^4] = \mathcal{O}(1)$ by assumption, so that $\max_i f(z_i)^4 + \mathbb{E}[e_i^4] = \mathcal{O}(1)$, also $\max_i (\vartheta' v_i)^2 \leq \max_i v_i' v_i = o(1)$, $\sum_{i=1}^n (\vartheta' v_i)^2 = \sum_{i=1}^n \vartheta' v_i v_i'$ $\vartheta = 1$, and $\operatorname{var}[\vartheta'\hat{b}]^{-2} \leq \max_i \sigma_i^{-2} = \mathcal{O}(1)$. $\square$

Proof of Theorem 4. We derive the limiting distribution of $\hat{\theta}$ with growing rank based on the following result regarding the joint normality of independent and not necessarily identical random variables as in Kline, Saggio, and Sølvsten (2020).

Let $\{q_{n,i}\}_{i,n}$ be a triangular array of row-wise independent random variables with

$\mathbb{E}[q_{n,i}] = 0$ and $\text{var}[q_{n,i}] = \sigma^2_{n,i}$, let $\{\dot{w}_{n,i}\}_{i,n}$ be a triangular array of non-random weights that satisfy $\sum_{i=1}^n \dot{w}_{n,i}\sigma^2_{n,i} = 1$ for $\forall n$, and let $(Q_n)_n$ be a sequence of symmetric non-random matrices in $\mathbb{R}^{n\times n}$ with zeros on the diagonal and having $2\sum_{i=1}^n \sum_{\ell\neq i} Q^2_{n,i\ell}\sigma^2_{n,i}\sigma^2_{n,\ell} = 1$. Define

$$\mathcal{S}_n := \sum_{i=1}^n \dot{w}_{n,i}q_{n,i}, \quad \mathcal{U}_n := \sum_{i=1}^n\sum_{\ell\neq i} Q_{n,i\ell}q_{n,i}q_{n,\ell}.$$

Lemma 3. *If* $\max_i \mathbb{E}[q^4_{n,i}] + \sigma^{-2}_{n,i} = O(1)$, *(i)* $\max_i \dot{w}^2_{n,i} = o(1)$, *and (ii) trace*$(Q^4_n) = o(1)$, *then* $(\mathcal{S}_n, \mathcal{U}_n)' \xrightarrow{d} \mathcal{N}(0, I_2)$.

Proof. See Appendix B in Kline, Saggio, and Sølvsten (2020) and Appendix A2 in Sølvsten (2020). □

The U-statistic representation holds because

$$\begin{aligned}
\hat{\theta} &= \sum_{i=1}^n y_i\tilde{w}_i'\hat{\gamma}_{-i} = \sum_{i=1}^n y_i\tilde{w}_i'\left(S_{ww}^{-1} - w_iw_i'\right)\sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n y_i\tilde{w}_i'\left(S_{ww}^{-1} + \frac{S_{ww}^{-1}w_iw_i'S_{ww}^{-1}}{1 - w_i'S_{ww}^{-1}w_i}\right)\sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n y_i\tilde{w}_i'\left(S_{ww}^{-1} + M_{W,ii}^{-1}S_{ww}^{-1}w_iw_i'S_{ww}^{-1}\right)\sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n y_iw_i'S_{ww}^{-1}\check{A}S_{ww}^{-1}\sum_{\ell\neq i} w_\ell y_\ell + \sum_{i=1}^n y_iw_i'S_{ww}^{-1}\check{A}M_{W,ii}^{-1}S_{ww}^{-1}w_iw_i'S_{ww}^{-1}\sum_{\ell\neq i} w_\ell y_\ell \\
&= \sum_{i=1}^n\sum_{\ell\neq i} y_iy_\ell w_i'S_{ww}^{-1}\check{A}S_{ww}^{-1}w_\ell + \sum_{i=1}^n\sum_{\ell\neq i} y_iy_\ell M_{W,ii}^{-1}w_i'S_{ww}^{-1}\check{A}S_{ww}^{-1}w_iw_i'S_{ww}^{-1}w_\ell \\
&= \sum_{i=1}^n\sum_{\ell\neq i} y_iy_\ell B_{W,i\ell} + y_iy_\ell M_{W,ii}^{-1}B_{W,ii}(1 - M_{W,i\ell}) = \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}y_iy_\ell.
\end{aligned}$$

Now, define $\check{e}_i := f(z_i) - p_k(z_i)'\alpha$ to be an approximation error (it is implicitly indexed by the unknown function but we omit this dependence for brevity). We can write the difference

$$\begin{aligned}
\hat{\theta} - \theta &= \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}y_iy_\ell - \sum_{i=1}^n \gamma'w_i\tilde{w}_i'\gamma \\
&= \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}\left(x_i'\beta + f(z_i) + e_i\right)\left(x_\ell'\beta + f(z_\ell) + e_\ell\right) - \sum_{i=1}^n \gamma'w_i\tilde{w}_i'\gamma \\
&= \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}\left(\gamma'w_i + e_i + \check{e}_i\right)\left(\gamma'w_\ell + e_\ell + \check{e}_\ell\right) - \sum_{i=1}^n \gamma'w_i\tilde{w}_i'\gamma \\
&= \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}\gamma'w_i\gamma'w_\ell + \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}(\gamma'w_ie_\ell + \gamma'w_\ell e_i) - \sum_{i=1}^n \gamma'w_i\tilde{w}_i'\gamma \\
&\quad + \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}(\gamma'w_i\check{e}_\ell + \gamma'w_\ell\check{e}_i + e_i\check{e}_\ell + e_\ell\check{e}_i) + \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}e_ie_\ell + \sum_{i=1}^n\sum_{\ell\neq i} C_{i\ell}\check{e}_i\check{e}_\ell
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{i=1}^{n} e_i \sum_{\ell \neq i} (\gamma' w_i + \gamma' w_\ell) C_{i\ell} + \sum_{i=1}^{n} \check{e}_i \sum_{\ell \neq i} (\gamma' w_i + \gamma' w_\ell + e_i + e_\ell) \\
&\quad + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} \check{e}_i \check{e}_\ell \\
&= \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma) e_i + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell + \mathcal{O}_p(nk^{-\alpha_f}),
\end{aligned}
$$

where the last equality follows from defining $\check{w}_i := \sum_{\ell=1}^{n} M_{W,i\ell} \frac{B_{W,\ell\ell}}{1-P_{W,\ell\ell}} w_\ell$, and the fact that by the Assumption 1

$$
\check{e}_i^2 \leq \left(f(z_i) - p_k(z_i)'\alpha\right)^2 = \mathbb{E}[\left(f(z_i) - p_k(z_i)'\alpha\right)^2] \leq \min_{\alpha \in \mathbb{R}^k} \mathbb{E}[\left(f(z_i) - p_k(z_i)'\alpha\right)^2] \leq Ck^{-2\alpha_f}.
$$

As $k \to \infty$, we have that

$$
\hat{\theta} - \theta = \sum_{i=1}^{n} \left(2\tilde{w}_i' \gamma - \check{w}_i' \gamma\right) e_i + \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell + o_p(1).
$$

Having dispensed with asymptotically negligible contributions to $\hat{\theta}$, asymptotic variance is

$$
\text{var}[\hat{\theta}] = \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma)^2 \sigma_i^2 + 2 \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} \sigma_i^2 \sigma_\ell^2,
$$

that is, a sum of two components given by

$$
\mathcal{V}_s := \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma)^2 \sigma_i^2, \quad \mathcal{V}_u := 2 \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} \sigma_i^2 \sigma_\ell^2.
$$

The normalized difference is then given by

$$
\text{var}[\hat{\theta}]^{-1/2} (\hat{\theta} - \theta) = \omega_1 \mathcal{S}_n + \omega_2 \mathcal{U}_n,
$$

with $\omega_1 := \text{var}[\hat{\theta}]^{-1/2} \mathcal{V}_s^{1/2}$, $\omega_2 := \text{var}[\hat{\theta}]^{-1/2} \mathcal{V}_u^{1/2}$, and

$$
\mathcal{S}_n := \mathcal{V}_s^{-1/2} \sum_{i=1}^{n} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma) e_i, \quad \mathcal{U}_n := \mathcal{V}_u^{-1/2} \sum_{i=1}^{n} \sum_{\ell \neq i} C_{i\ell} e_i e_\ell.
$$

Consider the case where the limit of ω_1 is nonzero. If it is not, then asymptotic normality of the difference $\text{var}[\hat{\theta}]^{-1/2}(\hat{\theta} - \theta)$ is implied by asymptotic normality of $\mathcal{U}_n$. Using notation of Lemma 3, we have that $\dot{w}_i = \mathcal{V}_s^{-1/2}(2\tilde{w}_i' \gamma - \check{w}_i' \gamma)$, and $Q_{i\ell} = \mathcal{V}_u^{-1/2} C_{i\ell}$.

To verify the condition *(i)* of Lemma 3, note that

$$
\begin{aligned}
\max_i \dot{w}_i^2 = \max_i \mathcal{V}_s^{-1} (2\tilde{w}_i' \gamma - \check{w}_i' \gamma)^2 &\leq \max_i 4\mathcal{V}_s^{-1} \left((\tilde{w}_i' \gamma)^2 + (\check{w}_i' \gamma)^2\right) \\
&= \max_i 4\omega_1^{-2} \frac{(\tilde{w}_i' \gamma)^2 + (\check{w}_i' \gamma)^2}{\text{var}[\hat{\theta}]} = o(1),
\end{aligned}
$$

where the last equality follows from Theorem 4 *(i)*, and the nonzero limit of ω_1.

For the condition *(ii)* of Lemma 3, denote $\tilde{A}_W := S_{ww}^{-1/2}\check{A}S_{ww}^{-1/2}$, and note that the first r eigenvalues of the $\tilde{A}_W$ matrix are equal to eigenvalues of the $\tilde{A}$ matrix. Now, with constants c_U and c_L not dependent on n, we have that $\text{trace}(C^4) \leq c_U \cdot \text{trace}(B_W^4) = c_U \cdot \text{trace}(\tilde{A}_W^4) \leq c_U\lambda_1^2 \cdot \text{trace}(\tilde{A}_W^2)$, and $\mathcal{V}_u \geq c_L \min_i \sigma_i^4 \cdot \text{trace}(\tilde{A}_W)$, which implies

$$\text{trace}(Q^4) \leq \frac{c_U\lambda_1^2 \cdot \text{trace}(\tilde{A}_W^2)}{\left(c_L \min_i \sigma_i^4 \cdot \text{trace}(\tilde{A}_W^2)\right)^2} = \mathcal{O}\left(\frac{\lambda_1^2}{\text{trace}(\tilde{A}_W^2)}\right) = o(1),$$

where the last equality follows from Theorem 4 *(ii)*. □

B. Additional Tables

Tables B1–B3 document the data and sample restrictions behind the estimation sample. Table B4 documents the firm controls that drive the richer specification, and Tables B5–B7 unpack the empirical specification ladder and its sensitivity to the series basis.

Table B1: Descriptive statistics of the cleaned worker–firm panel

	Mean	SD	p25	Median	p75
A. Worker-year panel					
Age	39.20	10.52	31.00	38.00	47.00
Months worked	8.19	1.69	8.00	9.00	9.00
Tenure (months)	103.57	103.76	22.00	69.00	153.00
Annual wage (k€)	13.92	10.46	7.99	10.53	16.02
Hourly wage (€)	7.17	5.17	4.15	5.33	7.99
Number of jobs	1.01	0.12	1.00	1.00	1.00
B. Worker composition and mobility					
Job-to-job moves (count)	1,803,534	–	–	–	–
Job-to-job movers (%)	9.85%	–	–	–	–
Female workers (%)	42.14%	–	–	–	–
Education: At most primary (%)	47.91%	–	–	–	–
Education: Secondary (%)	34.25%	–	–	–	–
Education: At least bachelor's (%)	17.52%	–	–	–	–
Education missing (%)	0.32%	–	–	–	–
Qualification: Specialized workers (%)	60.44%	–	–	–	–
Qualification: Generic workers (%)	15.27%	–	–	–	–
Qualification missing (%)	11.88%	–	–	–	–
Qualification: Top managers (%)	7.10%	–	–	–	–
Qualification: Middle managers (%)	5.31%	–	–	–	–
Lisboa (%)	35.13%	–	–	–	–
Norte (%)	35.06%	–	–	–	–
Centro (%)	18.28%	–	–	–	–
Alentejo (%)	4.54%	–	–	–	–
Algarve (%)	3.67%	–	–	–	–
Region (NUT2) other (%)	3.32%	–	–	–	–
C. Firm-year panel					
Workers	13.35	119.97	3.00	4.00	9.00
Fixed assets (M€)	0.72	26.38	0.01	0.03	0.14
Intermediate inputs (M€)	0.72	22.49	0.03	0.06	0.19
D. Panel counts					
Unique workers	3,532,497	–	–	–	–
Unique firms	484,704	–	–	–	–
Worker-year observations	18,314,740	–	–	–	–

Note: Descriptive moments are computed on the cleaned analysis sample (2008–2018). Panels A and C report distributional statistics for worker-year and firm-year variables, respectively. Panels B and D report composition, mobility, and sample-size counts. Percentages are sample shares in the corresponding panel.

Table B2: Missingness of estimation variables

Variable	Missing share
Age	0.00%
Education	0.32%
Firm ID	0.00%
Fixed assets	25.50%
Intermediate inputs	8.26%
Log hourly wage	0.01%
Qualification	11.88%
Gender	0.00%
Worker ID	0.00%
Workers	8.26%
Year	0.00%

Note: Entries report the share of worker-year observations in the cleaned analysis sample with missing values for each variable required by at least one empirical specification. The main empirical tables then restrict to non-missing controls, the largest connected component, and the leave-match-out set as reported in Table B3.

Table B3: Sample flow into the leave-match-out estimation set

Stage	*N*	Workers	Firms	Movers	Matches
Panel A: Parsimonious controls					
Loaded QP rows	23,304,646	4,217,716	533,360	1,758,967	7,104,038
Cleaned analysis sample	18,314,740	3,532,497	484,704	1,231,386	5,291,365
Non-missing controls	18,312,694	3,532,350	484,696	1,231,231	5,290,982
Largest connected component	17,256,654	3,282,116	333,180	1,212,893	5,021,308
Leave-match-out set	15,232,458	2,432,843	193,487	1,075,443	3,969,503
Panel B: Worker and firm-input controls					
Loaded QP rows	23,304,646	4,217,716	533,360	1,758,967	7,104,038
Cleaned analysis sample	18,314,740	3,532,497	484,704	1,231,386	5,291,365
Non-missing controls	11,892,426	2,642,775	296,549	663,784	3,487,931
Largest connected component	10,891,978	2,392,780	180,478	653,930	3,227,750
Leave-match-out set	9,439,305	1,719,643	105,442	578,409	2,457,963

Note: The table reports the empirical sample flow from loaded QP rows through the cleaned worker-year panel, non-missing controls, the largest worker-firm connected component, and the final leave-match-out set used for KSS estimation. The Panel A rows describe the larger parsimonious candidate sample; the main empirical tables use the Panel B leave-match-out set for both panels. Movers are workers observed at more than one firm, and matches are worker-firm pairs.

Table B4: Firm controls: within-firm time variation and missingness

	Non-missing firm-year	Firms (≥2y)	Mean years (firms ≥2y)	Within share (%)	Zero within SD (%)
A. Firm-control time variation in estimation sample					
Employment	2,397,757	383,679	5.99	7.88%	27.11%
Fixed assets	1,504,836	253,373	5.71	17.75%	5.99%
Intermediate inputs	1,874,310	290,932	6.22	9.75%	0.72%
B. Missingness by year (firm-year panel)					
Year	Firm-years	Workers miss.	Fixed assets miss.	Intermed. inputs miss.	All controls obs.
2008	253,005	0.00%	100.00%	26.27%	0.00%
2009	245,077	0.00%	100.00%	25.37%	0.00%
2010	223,267	0.00%	23.66%	23.66%	76.34%
2011	218,229	0.00%	23.01%	23.01%	76.99%
2012	202,744	0.00%	22.47%	22.47%	77.53%
2013	198,597	0.00%	21.20%	21.20%	78.80%
2014	202,625	0.00%	20.36%	20.36%	79.64%
2015	207,037	0.00%	19.89%	19.89%	80.11%
2016	211,892	0.00%	19.39%	19.39%	80.61%
2017	215,908	0.00%	18.80%	18.80%	81.20%
2018	219,376	0.00%	18.24%	18.24%	81.76%

Note: Panel A reports within-firm time variation of the Panel B firm controls in the cleaned firm-year panel (2008–2018), restricting to firms observed for at least two years; each control is measured on its own non-missing firm-years, so the row counts differ across controls and need not equal the joint estimation sample. The within-share statistic is the share of total variance on the estimation scale attributable to within-firm over-time variation: log employment and $\log(1+x)$ for fixed assets and intermediate inputs. Panel B reports year-by-year missingness rates; worker counts have zero missingness because employment is imputed from the matched employer–employee data, fixed assets are unavailable in 2008–2009 and remain about 18–24% missing thereafter, and intermediate inputs are about 18–26% missing throughout the sample. The final column reports the share of firm-years with workers, fixed assets, and intermediate inputs all observed.

Table B5: Stepwise changes in wage variance decomposition

Change	$\Delta\sigma^2_\alpha$	$\Delta\sigma^2_\psi$	Δ cov(α, ψ)
Panel A: Parsimonious controls (Panel B sample)			
Add linear observables	-0.0193	-0.0068	-0.0069
Add heterogeneity at degree 1	0.0061	-0.0001	0.0011
Add nonlinearity without interactions	0.0171	0.0003	0.0025
Add heterogeneity in nonlinear basis	0.0118	-0.0006	0.0011
Add nonlinearity with interactions	0.0228	-0.0002	0.0025
Panel B: Worker and firm-input controls			
Add linear observables	-0.0754	-0.0206	-0.0355
Add heterogeneity at degree 1	-0.0232	-0.0015	-0.0020
Add nonlinearity without interactions	0.0169	-0.0019	-0.0051
Add heterogeneity in nonlinear basis	-0.0020	-0.0039	0.0016
Add nonlinearity with interactions	0.0381	-0.0043	-0.0015

Note: Entries report within-panel differences in the bias-corrected variance components when moving from one specification to the next on the reported leave-match-out estimation sample. The transitions branch from the linear additive specification: one path adds heterogeneity first, the other adds nonlinearity first, and both arrive at the full model. These are descriptive model-comparison deltas, not formal tests.

Table B6: Alternative series bases and the wage variance decomposition

Basis	Nonlinear specification	σ^2_α	σ^2_ψ	cov(α, ψ)	var$(\tilde{y})$ / var(y)
Panel A: Parsimonious controls (Panel B sample)					
Degree-5 polynomial	Common	0.5509	0.1294	0.0841	0.9783
Degree-5 polynomial	Group-specific	0.5627	0.1288	0.0852	0.9913
Degree-5 Hermite	Common	0.5445	0.1295	0.0836	0.9709
Degree-5 Hermite	Group-specific	0.5499	0.1288	0.0842	0.9765
Cubic B-spline	Common	0.5535	0.1297	0.0841	0.9808
Cubic B-spline	Group-specific	0.5513	0.1290	0.0834	0.9764
Panel B: Worker and firm-input controls					
Degree-5 polynomial	Common	0.4947	0.1134	0.0478	0.8319
Degree-5 polynomial	Group-specific	0.4926	0.1095	0.0495	0.8271
Degree-5 Hermite	Common	0.4934	0.1134	0.0478	0.8306
Degree-5 Hermite	Group-specific	0.4934	0.1095	0.0499	0.8286
Cubic B-spline	Common	0.4963	0.1096	0.0467	0.8276
Cubic B-spline	Group-specific	0.4765	0.1073	0.0484	0.8070

Note: Entries report residualized leave-match-out KSS variance components and the residualized-wage variance ratio on the common estimation sample. "Common" denotes the nonlinear homogeneous specification; "Group-specific" denotes the full specification that interacts the basis with worker groups. The polynomial and Hermite specifications use complete degree-5 multivariate bases. Although these bases span the same polynomial space and fit the same wage, their variance decompositions need not coincide, because the two families place their constant and low-order terms differently relative to the fixed-effect normalization; the comparison therefore also measures that normalization sensitivity. The B-spline specification is additive across continuous inputs and uses cubic bases with quantile-spaced knots. Spline counts match the polynomial model's realized control dimension. The table reports point estimates only.

Table B7: Control dimension and specification diagnostics

Model	k
Panel A: Parsimonious controls (Panel B sample)	
AKM baseline	0
Linear additive controls	2
Heterogeneous linear controls	3
Nonlinear homogeneous controls	4
Full model	9
Panel B: Worker and firm-input controls	
AKM baseline	0
Linear additive controls	10
Heterogeneous linear controls	102
Nonlinear homogeneous controls	129
Full model	3004

Note: k is the number of control columns partialled out before the KSS decomposition. Sample-flow counts are reported separately in Table B3.